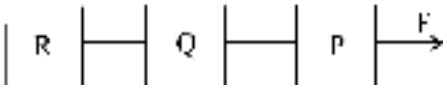
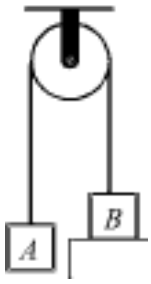
PHYSICS

1) Three equal masses P, Q and R are pulled with a constant force F. They are connected to each other with strings. The ratio of the tension between PQ and QR is :-



(1) 1 : 1
(2) 2 : 1
(3) 1 : 2
(4) 3 : 1

2) In the diagram shown; block A of mass 2 kg is hanging from the string passing over a smooth pulley and block B is placed on the top of a table. If the reaction of the table is 10 N, mass of block B is : (g=10m/s^2)



(1) 1 Kg
(2) 2 Kg
(3) 3 Kg
(4) 4 Kg

3)

Choose the **Incorrect** option :-

(1) if $\hat{a} \cdot \hat{v} = 1$ then object is moving with increasing speed without change in direction
(2) if $\hat{a} \cdot \hat{v} = -1$ then object definitely changes its direction
(3) if $\hat{r} \cdot \hat{v} = -1$ then object is moving towards origin
(4) if $\hat{r} \cdot \hat{v} = -1$ then object is moving away from origin

4) A particle is moving in a straight line with initial velocity u and uniform acceleration f. If the sum of the distances travelled in t^{th} and $(t + 1)^{th}$ seconds is 100cm, then its velocity after t seconds, in cm/s, is

(1) 20
(2) 30
(3) 50

(4) 80

5) A body is projected upward with velocity 20 m/s at an angle of 30° with the horizontal from a height 40 m. The time after which body reaches the ground :-

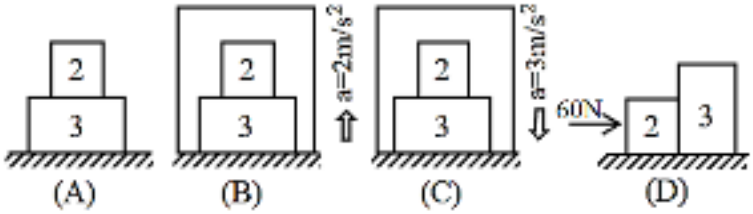
(1) 1 sec

(2) 2 sec

(3) 3 sec

(4) 4 sec

6) In which case normal reaction between the blocks will be least :



(1) A

(2) B

(3) C

(4) D

7) Percentage error in measuring the radius and mass of a solid sphere are 1% and 2% respectively. The error in measurement of density :-

(1) 9%

(2) 6%

(3) 5%

(4) 4%

8) A particle is moving along a circular path of radius 10 m and with uniform speed 10 m/s. What will be the average acceleration when the particle completes half revolution.

(1) zero

(2) 20 ms^{-2}

(3) $20\pi\ ms^{-2}$

(4) $\frac{20}{\pi} ms^{-2}$

9) A bullet is fired from a gun. The force on the bullet is given by: $F = 600 - 2 \times 10^5 t$.
Where F is in newton and t in second. The force on the bullet becomes zero as soon as it leaves the barrel. What is the average impulse imparted to the bullet ?

(1) 9 N-s

(2) Zero

(3) 0.9 N-s

(4) 1.8 N-s

10) The initial speed of a projectile fired from ground is u. At the highest point during its motion, the speed of projectile is $\frac{\sqrt{3}}{2}u.$ The time of flight of the projectile is:

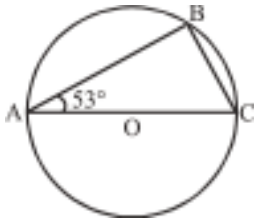
(1) $\frac{u}{2g}$

(2) $\frac{u}{g}$

(3) $\frac{2u}{g}$

(4) $\frac{\sqrt{3}u}{g}$

11) A particle moves along a straight line from A to B and then from B to C. If AC is the diameter of the circle, the distance covered by the particle is:-



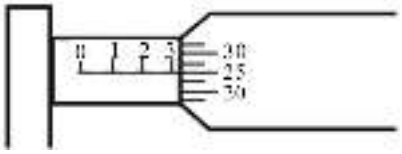
(1) 14r/5

(2) 2r/3

(3) 2r

(4) 10r/3

12) Find the reading of screwguage. Circular scale has 100 divisions. (Pitch = 1 mm)



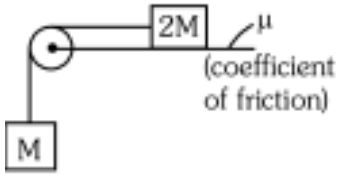
(1) 3.25 mm

(2) 3.24 mm

(3) 3.30 mm

(4) 3.41 mm

13) In the given diagram acceleration of the system is: (Consider string as light & inextensible and pulley as light and frictionless)



(1) g/3

(2) $\frac{g(1-2\mu)}{3}$

(3) $\frac{g(1-\mu)}{2}$

(4) zero

14) A train takes 4 min. to go between stations 2.25 km apart starting and finishing at rest. The acceleration is uniform for the first 40 sec and the deceleration is uniform for the last 20 sec. Assuming the velocity to be constant for the remaining time, then maximum speed of the train is :-

(1) 75 m/sec

(2) 18.75 m/sec

(3) 37.5 m/sec

(4) 10.7 m/sec

15) A ship A is moving Westwards with a speed of 10 km h^{-1} and a ship B 100 km South of A, is moving Northwards with a speed of 10 km h^{-1}. The time after which the distance between them becomes shortest, is :-

(1) 5 h

(2) $5\sqrt{2}$h

(3) $10\sqrt{2}$h

(4) 0 h

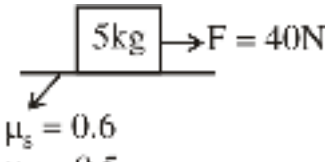
16) A ball is thrown vertically upwards with a certain velocity. Neglecting air resistance, identify the wrong statement(s) regarding the motion of the ball.
(A) During the upward motion net force on the ball acts upwards.
(B) During the downward motion net force on the ball acts downwards.
(C) At the highest position no force acts on the ball.
(D) The net force on the ball always act in the downward direction.

(1) A, B

(2) B, C

(3) B, D

(4) A, C



17) Match the column-I with column-II.

	Column-I		**Column-II**
(P)	a (m/s^2)	(1)	25
(Q)	N (Newton)	(2)	30
(R)	f_L (Newton)	(3)	50
(S)	f_k (Newton)	(4)	3

(1) P→3, Q→4, R→2, S→1

(2) P→4, Q→3, R→2, S→1

(3) P→1, Q→2, R→3, S→4

(4) P→4, Q→2, R→1, S→3

18) What is the projection of $3\hat{i} + 4\hat{k}$ on the y-axis ?

(1) 3

(2) 4

(3) 5

(4) zero

19)

A train 100 m long travelling at 40 ms^{-1} overtakes another train 200 m long travelling at 30 ms^{-1}. The time taken by the first train to pass the second train is :-

(1) 30 s

(2) 40 s

(3) 50 s

(4) 60 s

20) A number of bullets are fired in all possible directions with the same initial velocity u. The maximum area of ground covered by bullets is:

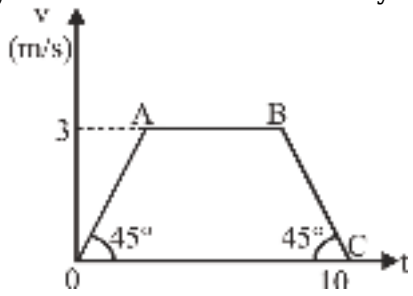
(1) $\pi\left(\frac{2u^2}{g}\right)^2$

(2) $3\pi\left(\frac{u}{g}\right)^2$

(3) $5\pi\left(\frac{u}{2g}\right)^2$

(4) $\pi\left(\frac{u^2}{g}\right)^2$

21) For the v-t graph shown in figure total time of journey is 10 s. If maximum velocity during the motion is 3 m/s, the time for which motion is uniform is :-



(1) 2s

(2) 4s

(3) 6s

(4) 8s

22)

What is the value of $\left(\vec{A}+\vec{B}\right)\bullet\left(\vec{A}\times\vec{B}\right)$?

(1) 0

(2) $A^2 - B^2$

(3) $A^2 + B^2 + 2AB$

(4) None of these

23) If the equation of path of a projectile is given by $y = 12x - \frac{3}{4}x^2$ then find horizontal range :-

(1) 12.4 m

(2) 21 m

(3) 16 m

(4) 36 m

24) If energy(E), velocity(V) and time(T) are considered as fundamental quantities then dimension of mass will be :-

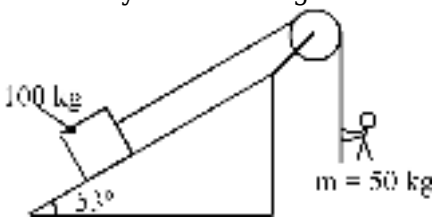
(1) $[E\ V^{-2}\ T^0]$

(2) $[E^2\ V^{-2}\ T^0]$

(3) $[E\ V^2\ T]$

(4) $[E^{-1}\ V^2\ T^1]$

25) In the given figure, by what acceleration the boy must go up so that 100 kg block remains stationary on the wedge ? The wedge is fixed and friction is absent everywhere ($g = 10\ m/s^2$)



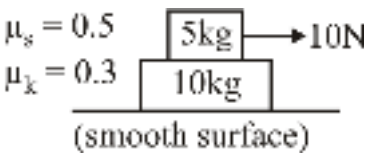
(1) $2m/s^2$

(2) $4m/s^2$

(3) $6m/s^2$

(4) $8m/s^2$

26) Find friction force between 5 kg and 10 kg ?



(1) $\frac{10}{3}$N

(2) $\frac{20}{3}$N

(3) 25N

(4) 15N

27) **Assertion (A) :-** Magnitude of contact force is always greater than the magnitude of non-zero friction force.
Reason (R) :- Contact force is the resultant of the friction force & normal reaction.

(1) Both Assertion & Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion & Reason are true but Reason is not correct explanation of Assertion.

(3) Assertion is true but Reason is false

(4) Both Assertion & Reason are false

28) Starting from rest a lift is accelerating upwards with acceleration 2.2 m/s^2. Two seconds after the start of lift an object is thrown upward with speed 18 m/s w.r.t. lift. Find the time taken by object to again hit the floor of lift. Assume lift to be open roof and g= 9.8 m/s^2 :-

(1) 10 sec

(2) 6 sec

(3) 12 sec

(4) 3 sec

29) A particle moves along a straight line such that in different cases, its velocity is changing with time as shown in List–I. List–II represents value of position. If at t = 0 particle was at x = 0, then match the velocity equation in list-I with their corresponding position values at t = 1 sec in list-II :-

List-I		**List-II**	
(I)	$v = \frac{t^2}{2} + 1$	(P)	36
(II)	$v = 3t^3$	(Q)	2
(III)	$v = 2t$	(R)	$\frac{3}{4}$
(IV)	$v = 2t + 2$	(S)	$\frac{7}{6}$
		(T)	3
		(U)	1

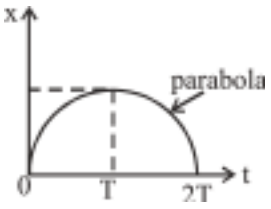
(1) I-S;II-R;III-R;IV-T

(2) I-R;II-S;III-U;IV-T

(3) I-S;II-R;III-T;IV-U

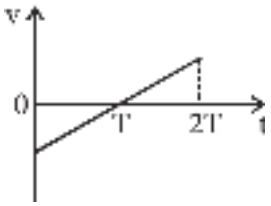
(4) I-S;II-R;III-U;IV-T

30) The x–t graph of a particle moving along a straight line is shown in figure.

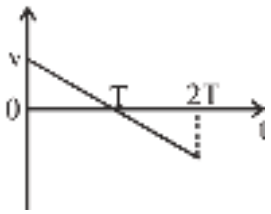


The v–t graph of the particle is correctly shown by

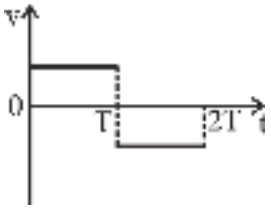
(1)



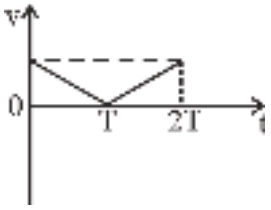
(2)



(3)



(4)



31) The values of x and y for which vectors $\vec{A} = \left(6\hat{i} + x\hat{j} - 2\hat{k}\right)$ and $\vec{B} = \left(5\hat{i} - 6\hat{j} - y\hat{k}\right)$ may be parallel are :

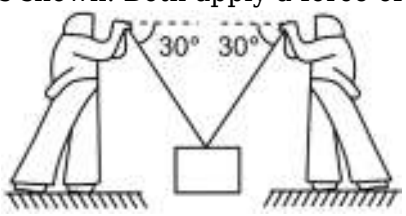
(1) $x = 0, y = \frac{2}{3}$

(2) $x = -\frac{36}{5}, y = \frac{5}{3}$

(3) $x = -\frac{15}{3}, y = \frac{23}{5}$

(4) $x = \frac{36}{5}, y = \frac{15}{4}$

32) Two men pull a rope from which a mass is hanging as shown. Both apply a force of 100 N each.



What the force needed so that rope become horizontal ?

(1) 200 N

(2) 400 N

(3) 600 N

(4) Rope cannot become horizontal

33) A man of mass 60 kg slides down a pole. He is pressing the pole with a force of 600 N. The coefficient of friction between the hands and the pole is 0.5, with what acceleration will the man slide down ($g = 10\ m/s^2$) :-

(1) $1\ m/s^2$

(2) $2.5\ m/s^2$

(3) $10\ m/s^2$

(4) $5\ m/s^2$

34) A particle moves in a straight line with velocity V for time t_o. Then its velocity becomes 2 V for next t_o time. Finally its velocity becomes 3 V for time T. If its average speed during complete journey was 2.5 V, then value of T is :-

(1) t_o

(2) $2\ t_o$

(3) $3\ t_o$

(4) $4\ t_o$

35)

The least count of a stop watch is 0.1 sec. The time of 20 oscillations of the pendulum is found to be 20 sec. The percentage error in the time period is

(1) 0.25%

(2) 0.5%

(3) 0.75%

(4) 1.0%

36) The position of a particle moving along a straight line is given by $x(t) = \frac{A}{B}(1-e^{-At})$ where A and B are constants. The dimensions of $\frac{A^3}{B}$ is same as

(1) linear momentum

(2) moment of inertia

(3) relative velocity

(4) acceleration

37) Find area of ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

(1) 6π

(2) $36\ \pi$

(3) $\frac{3\pi}{2}$

(4) $\frac{9\pi}{4}$

38) In a certain system, the units of length, mass and time are chosen to be 1 cm, 100 g and 10 s. The unit of kinetic energy will be:

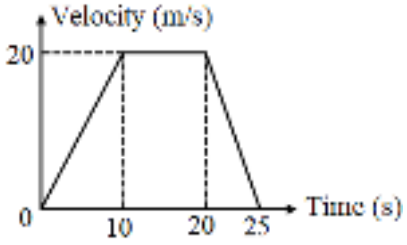
(1) 0.1 erg

(2) 1 erg

(3) 10 erg

(4) None of these

39) The velocity-time graph of a car moving along a straight road is shown in figure. The average velocity of the car in first 25 seconds is



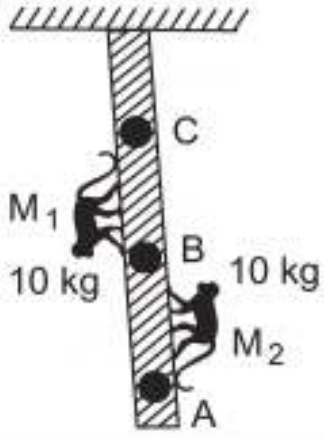
(1) 20 m/s

(2) 14 m/s

(3) 10 m/s

(4) 17.5 m/s

40) If monkey M_1 is at rest and breaking strength of string is 260 N, then α_{max} of M_2 with which it can accelerate, so that string will not break is :



(1) 8 m/sec^2 ↑

(2) 6 m/sec^2 ↑

(3) 4 m/sec^2 ↑

(4) None

41) Two persons A and B are sitting in two cars 1 and 2 such that the velocity of the cars is given by

$\vec{v}_1 = 3t^2\hat{i} - \frac{4}{3}t^6\hat{j}$ and $\vec{v}_2 = 3\left(t^2 - 5t\right)\hat{i}$ respectively. The fictitious (pseudo) force on B as observed by A at t = 1 sec will be :- (m_A = 70 kg, m_B = 50 kg)

(1) $70\left(4\hat{j} - 3\hat{i}\right)$ N

(2) $100\left(4\hat{j} - 3\hat{i}\right)$ N

(3) $50\left(8\hat{j} - 15\hat{i}\right)$ N

(4) $70\left(8\hat{j} - 15\hat{i}\right)$ N

42) The diameter of a cylinder is measured using a vernier calliper with no zero error. It is found that the zero of the Vernier scale lies between 5.10 and 5.15 cm of the main scale. The vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the Vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is

(1) 4.112 cm

(2) 5.124 cm

(3) 5.136 cm

(4) 5.148 cm

43) A body of mass 6 kg moves in a straight line according to the equation $x = t^3 - 75t$ where x denotes the distance in metre and t the time in second. The force on the body at t = 4 second is :–

(1) 64 newton

(2) 72 newton

(3) 144 newton

(4) 36 newton

44) A vernier callipers has 20 division on the vernier scale which coincide with 19 divisions of main scale, one division of main scale is 0.5 mm then least count of vernier callipers will be :-

(1) 0.035 mm

(2) 0.025 mm

(3) 0.25 mm

(4) 0.40 mm

45) A stone is projected horizontally from the top of a tower with a speed of 20 m/s. It strikes the ground at a horizontal distance of 400 m. The time of flight of the stone is :-

(1) 10 s

(2) 20 s

(3) 30 s

(4) 40 s

CHEMISTRY

1) In a reversible reaction, two substances are in equilibrium. If the concentration of each one is doubled, the equilibrium constant will be

(1) Reduced to half its original value

(2) Reduced to one fourth of its original value

(3) Doubled

(4) Constant

2)

The equilibrium constant for the reaction, $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 4×10^{-4} at 2000 K. In presence of a catalyst, equilibrium is attained ten times faster. Therefore, the equilibrium constant, in presence of the catalyst, at 2000 K is:

(1) 40×10^{-4}

(2) 4×10^{-4}

(3) 4×10^{-3}

(4) difficult to compute without more data.

3) Partial pressure of O_2 in the reaction $2Ag_2O(s) \rightleftharpoons 4\,Ag(s) + O_2(g)$ is

(1) K_p

(2) $\sqrt{K_p}$

(3) $\sqrt[3]{K_p}$

(4) $2K_p$

4) A quantity of PCl_5 was heated in a 10 dm^3 vessel at 250°C : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. At equilibrium, the vessel contains 0.1 mole of PCl_5 and 0.2 mole of Cl_2. The equilibrium constant of the reaction is

(1) 0.04

(2) 0.025

(3) 0.02

(4) 0.05

5) $A + B \rightleftharpoons C + D$. If initially the concentrations of A and B are equal but at equilibrium, concentration of D will be twice that of A, then what will be the equilibrium constant of the reaction ?

(1) $\frac{4}{9}$

(2) $\frac{9}{4}$

(3) $\frac{1}{9}$

(4) 4

6) Which of the following is true-

(1) Active mass of pure liquids and solids are taken unity.

(2) Active mass of gases can be expressed is terms of atm.

(3) Law of mass action is applicable on reversible reaction

(4) All of above

7) The equilibrium constant (K_c) for the reaction $HA + B \rightleftharpoons BH^+ + A^-$ is 100. If the rate constant for the forward reaction is 10^5, then rate constant for the backward reaction is

(1) 10^7

(2) 10^3

(3) 10^{-3}

(4) 10^{-5}

8) **Statement-1 :-** In exothermic reaction, heat is evolved during reaction.
Statement-2 :- Product is more stable then reactant.

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

9) The aqueous solution of following salt will have the lowest pH

(1) NaClO

(2) $NaClO_4$

(3) $NaClO_3$

(4) $NaClO_2$

10) pK_a of weak acid (HA) and pK_b of weak base (BOH) are 3.2 and 3.4 respectively. The pH of their salt (BA) solution is

(1) 7.0

(2) 1.0

(3) 7.2

(4) 6.9

11) The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T(K). The correct option for the pH of dimethylammonium acetate solution is

(1) 8.50

(2) 5.50

(3) 7.75

(4) 6.25

12)

Match the entries of column I with appropriate entries of column II and choose the correct option.

	Column-I		**Column-II**
(A)	pK_b of X^- (K_a of HX = 10^{-6})	(p)	6.96
(B)	pH of 10^{-8} M HCl	(q)	8
(C)	pH of 10^{-2} M acetic acid solution (Take K_a of acetic acid = 2×10^{-5})	(r)	3.3
(D)	pH of a solution obtained by mixing equal volumes of solutions with pH 3 and 5.	(s)	3.35

(1) A-q, B-p, C-r, D-s

(2) A-r, B-s, C-q, D-p

(3) A-q, B-p, C-s, D-r

(4) A-p, B-q, C-s, D-r

13) What will be the pH of pure water at 60°C ?

(1) Equal to 7.0

(2) Greater than 7.0

(3) Less than 7.0

(4) Equal to zero

14)

The K_a values of acetic acid, hypochlorous acid and formic acid are 1.74×10^{-5}, 3.0×10^{-8} and 1.8×10^{-4} respectively Which of the following orders of pH of 0.1 mol dm^{-3} solutions of these acids is correct ?

(1) acetic acid > hypochlorous acid > formic acid

(2) hypochlorous acid > acetic acid > formic acid

(3) formic and > hypochlorous acid > acetic acid

(4) formic acid > acetic acid > hypochlorous acid

15) K_a for CH_3COOH is 1.8×10^{-5} and K_b for NH_4OH is 1.8×10^{-5}. The pH of ammonium acetate will be

(1) 7.005

(2) 4.75

(3) 7.0

(4) Between 6 and 7

16) **Assertion (A):** The ionisation of hydrogen sulphite in water is low in the presence of hydrochloric acid.
Reason (R): Hydrogen sulphite is a weak acid.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false

(4) Both A and R are false

17) **Assertion (A):** Aqueous solution of ammonium carbonate is basic.
Reason (R): Acidic/basic nature of a salt solution of a salt of weak acid and weak base depends on K_a and K_b value of the acid and the base forming it.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) Both A and R are false.

18) At 25°C, the dissociation constant of a base, BOH is 1.0×10^{-12}. The concentration of hydroxyl ions in 0.01 M aqueous solution of the base would be:

(1) 1.0×10^{-6} mol L^{-1}

(2) 1.0×10^{-7} mol L^{-1}

(3) 2.0×10^{-6} mol L^{-1}

(4) 1.0×10^{-5} mol L^{-1}

19) Aqueous solution of which of the following compounds is the best conductor of electric current ?

(1) Ammonia, NH_3

(2) Fructose, $C_6H_{12}O_6$

(3) Acetic acid, $C_2H_4O_2$

(4) Hydrochloric acid HCl

20) What is the conjugate base of OH^- ?

(1) O_2

(2) H_2O

(3) O^-

(4) O^{2-}

21) In which one of the following changes, there is transfer of five electrons?

(1) $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

(2) $CrO_4^{2-} \rightarrow Cr^{3+}$

(3) $MnO_4^- \rightarrow MnO_2$

(4) $MnO_4^- \rightarrow Mn^{2+}$

22) In the following reaction,
$4P + 3KOH + 3H_2O \rightarrow 3KH_2PO_2 + PH_3$

(1) P is oxidised only

(2) P is oxidised as well as reduced

(3) P is reduced only

(4) none of the above

23) The oxidation state of Cr in $[Cr(NH_3)_4Cl_2]^+$ is:

(1) +1

(2) +2

(3) +3

(4) 0

24) For the redox reaction,
$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$
the correct coefficients of the reactants for the balanced equation are:

	MnO_4^-	$C_2O_4^{2-}$	H^+
(1)	5	16	2
(2)	2	5	16
(3)	2	16	5
(4)	16	5	2

(1) 1

(2) 2

(3) 3

(4) 4

25) The value of n in the following equation is:
$Cr_2O_7^{2-} + 14H^+ + nFe^{2+} \rightarrow 2Cr^{3+} + nFe^{3+} + 7H_2O$

(1) 7

(2) 6

(3) 2

(4) 3

26) Which of the following shows nitrogen with its increasing order of oxidation number ?

(1) $NO < N_2O < NO_2 < NO_3^- < NH_4^+$

(2) $NH_4^+ < N_2O < NO < NO_2 < NO_3^-$

(3) $NH_4^+ < N_2O < NO_2 < NO_3^- < NO$

(4) $NH_4^+ < NO < N_2O < NO_2 < NO_3^-$

27) What are the stoichiometric coefficient of the following reaction ?

$pI_2 + qOH^- \rightarrow IO_3^- + rI^- + 3H_2O$

(1) p → 6, q → 3, r → 5

(2) p → 3, q → 2, r → 3

(3) p → 3, q → 3, r → 3

(4) p → 3, q → 6, r → 5

28) In $_{92}U^{235}$ nucleus, the number of electrons is :-

(1) 92

(2) 235

(3) 143

(4) Zero

29) Ratio of mass of neutron and proton is :-

(1) Less than one

(2) More than one

(3) Exactly equal to one

(4) Can't be predicted

30) Rutherford's experiment on scattering of alpha particles showed for the first time that atom has:

(1) Electrons

(2) Protons

(3) Nucleus

(4) Neutrons

31) Which of the following conclusions regarding the structure of atom is true based on Rutherford's α-particle scattering experiment ?

(1) The positive charge is concentrated in a very small volume of the atom

(2) The positive charge is scattered with the electrons throughout the atom

(3) The volume occupied by the nucleus is half of the volume of atom

(4) Most of the space in the atom is occupied by the neutrons.

32) Angular momentum for P–shell electron is :–

(1) $\frac{3h}{\pi}$

(2) Zero

(3) $\frac{\sqrt{2}h}{2\pi}$

(4) None

33) The mass no. of an element is 39. The number of neutrons in its nucleus is one more than the number of protons. The number of protons, neutrons and electrons respectively in its atom would be

(1) 19, 20, 19

(2) 19, 19, 20

(3) 20, 19, 19

(4) 20, 19, 20

34) Energy of e^- in 2nd orbit of H-atom is-

(1) 13.6 eV

(2) –3.4 eV

(3) –1.51 eV

(4) –0.54 eV

35) If the K.E. of e^- in first and second orbit of Li^{+2} ion is x and y respectively then :

(1) x < y

(2) x > y

(3) x = y

(4) None of these

36) Find out ratio of following
$(\nu_{max})_{Lyman} : (\nu_{max})_{Brackett}$

(1) 1 : 16

(2) 16 : 1

(3) 4 : 1

(4) 1 : 4

37) Moles of O_3 in its 12.044×10^{22} molecules of O_3 will be :

(1) 2 mol

(2) 0.4 mol

(3) 1 mol

(4) 0.2 mol

38) If 5 moles of A is mixed 15 moles of B then what is the mole fraction of A in the mixture ?

(1) 0.75

(2) 0.35

(3) 0.25

(4) 0.5

39) Volume of $H_2(g)$ required to react completely with 3L of $N_2(g)$ to form ammonia at same T & P condition will be :

(1) 7 L
(2) 6 L
(3) 9 L
(4) 4 L

40) When 5 moles of A will react with 3 moles of B according to given balanced reaction

$$2A + 6B \rightarrow C$$

Then which of the following reactant will be in excess amount

(1) A
(2) B
(3) can't be determined
(4) Both will act as limiting reagent

41) % of H in methane will be :

(1) 25%
(2) 30%
(3) 17%
(4) 32%

42) The no. of atoms of A and B in their molecule are 9×10^{20} and 3×10^{20} respectively then their empirical formula will be :

(1) AB_3
(2) A_3B
(3) AB
(4) A_9B

43) 2 moles of urea is dissolved in 20 kg of solvent then molality of the solution formed is :

(1) 0.1 m
(2) 0.2 m
(3) 1 m
(4) 1.5 m

44) The ratio of the energy of a photon of wavelength 3000 Å to that of a photon of wavelength 6000 Å is :

(1) $\frac{1}{2}$
(2) 2

(3) 3

(4) $\frac{1}{3}$

45) According to Bohr's theory, the angular momentum to an electron in 5th orbit is :

(1) 25 (h/π)

(2) 1.0 (h/π)

(3) 10 (h/π)

(4) 2.5 (h/π)

BIOLOGY

1) The secretion of which gland helps in the lubrication of penis?

(1) Seminal vesicle

(2) Prostate

(3) Bulbourethral

(4) Epididymis

2) Which of the following describes the function of the skin bag which encloses the testis ?

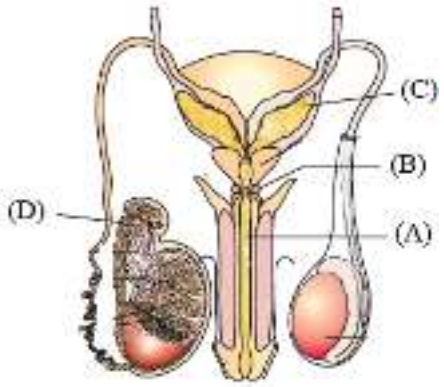
(1) It synthesises and secretes testicular hormones called androgens.

(2) It provides nutrition to the germ cells.

(3) It maintains the low temperature of the testes necessary for spermatogenesis.

(4) It is made up of a special tissue that helps in erection of the penis to facilitate insemination.

3) Given below is the diagramatic view of male reproductive system. Identify correct labelling for A, B, C and D ?



Option :-

	(A)	(B)	(C)	(D)
(1)	Penis	Prostate	Seminal Vesicle	Vas deferens
(2)	Urethra	Bulbourethral gland	Urinary bladder	Epididymis
(3)	Urethra	Bulbourethral gland	Seminal Vesicle	Epididymis
(4)	Penis	Bulbourethral gland	Seminal Vesicle	Vas deferens

(1) 1

(2) 2

(3) 3

(4) 4

4) Which one is correct sequence of pathway of movement of sperm :-

(1) Rete testis, epididymis, vasa efferentia, urethra

(2) Rete testis, urethra, vasa efferentia, epididymis

(3) Urethra, epididymis, vasa efferentia, rete testis

(4) Rete testis, epididymis, vasa deferens, urethra

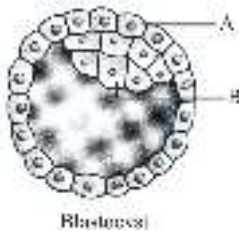
5) Relaxin hormone is secreted by.
(A) Placenta
(B) Secondary follicle
(C) Graafian follicle
(D) Corpus luteum

(1) A and B

(2) A and D

(3) B and C

(4) B and D



6) Identify A and B and their respective functions:

	(A)	(B)	Function of	
			(A)	(B)
(1)	Trophoblast	Inner cell mass	Attach to the endometrium	Differentiated as embryo
(2)	Trophoblast	Inner cell mass	Differentiated as embryo	Attach to the Endometrium
(3)	Mesoderm	Inner cell mass	Differentiated as embryo	Attach to the Endometrium
(4)	Ectoderm	Mesoderm	Attach to the endometrium	Differentiated as embryo

(1) 1

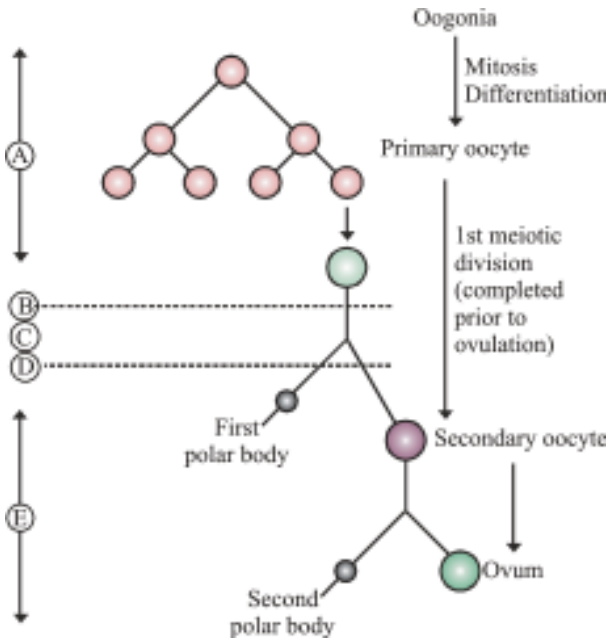
(2) 2

(3) 3

(4) 4

7)

The following refers to schematic representation of oogenesis.



Identify A to E correctly.

(1) (A) → Fetal life, (B) → Birth, (C) → Puberty, (D) → Adult reproductive life, (E) → Child hood

(2) (A) → Fetal life, (B) → Birth, (C) → Child hood, (D) → Puberty, (E) → Adult reproductive life

(3) (A) → Adult reproductive life, (B) → Birth, (C) → Puberty, (D) → Child hood, (E) → Fetal life

(4) (A) → Birth, (B) → Child hood, (C) → Fetal life, (D) → Puberty, (E) → Adult reproductive life

8)

Arrange following events of female reproductive cycle (human) in sequence :-
(I) Secretion of FSH
(II) Growth of corpus luteum
(III) Growth of follicles
(IV) Ovulation

(V) LH Surge

(1) III → I → IV → II → V

(2) I → III → V → IV → II

(3) I → IV → III → V → II

(4) II → I → III → IV → V

9) Consider the following four statements (a-d) and select the option which includes all the **incorrect** ones only.
(a) Each testicular lobule contains 750 highly coiled seminiferous tubules, in which sperms are produced.
(b) Sertoli cells undergo meiotic divisions finally leading to sperm formation.
(c) The regions outside the seminiferous tubules called interstitial spaces, contain small blood vessels, Leydig cells, and some immunologically competent cells.
(d) The scrotum helps in maintaining the low temperature of the testes, necessary for spermatogenesis.

(1) Statements (b), (c) and (d).

(2) Statements (a), (b).

(3) Statements (a), (b) and (c).

(4) Statements (a), (c) and (d).

10) Consider the following statements and select the wrong one :-

(1) Parturition is induced by a complex neuro endocrine mechanism.

(2) The first movement of the foetus & appearance of hairs on the head are usually observed during fifth month of pregnancy.

(3) During parturition, secretion of progesterone decreases but secretion of estrogen increases from placenta.

(4) Progesterone and estrogen are produced in women only during pregnancy.

11)

Match the column–A and B about the embryonic development of human :-

Column-A	Column-B
(i) End of one month	(a) Most of the major organ systems develope
(ii) End of second month	(b) Appearance of hair on the head
(iii) End of three month	(c) Heart formation
(iv) During fifth month	(d) Eye-lids separated
(v) End of sixth month	(e) Limbs and digits formation

(1) (i) – e, (ii) – b, (iii) – c, (iv) – a, (v) – d

(2) (i) – e, (ii) – c, (iii) – b, (iv) – b, (v) – d

(3) (i) – c, (ii) – e, (iii) – b, (iv) – a, (v) – d

(4) (i) – c, (ii) – e, (iii) – a, (iv) – b, (v) – d

12)

Consider the following structures.

Testes, scrotum, epididymis, vas deferens, seminal vesicles, urinary bladder, prostate gland, ovaries, fallopian tubes, Bartholin glands How many above structure are found only in male as paired structure ?

(1) 10

(2) 6

(3) 4

(4) 7

13) Match the following :-

	Column I		Column II
A	Menopause	i	Leydig Cells
B	Fertilization	ii	Parturition
C	Androgens	iii	50 years of age

D	Oxytocin	iv	Ampullary region of fallopian tube

(1) A - iii, B - iv, C - ii, D - i
(2) A - iii, B - iv, C - i, D - ii
(3) A - iv, B - iii, C - i, D - ii
(4) A - iv, B - iii, C - ii, D - i

14) After ovulation Graafian follicle becomes -

(1) Secondary follicle
(2) Corpus albicans
(3) Corpus luteum
(4) Primary follicle

15) Functional mammary gland is the characteristic feature of :-

(1) All female animals
(2) All mammals
(3) All mammalian female
(4) only prototherian female

16) Thickness of uterine wall maximum will be, on which day of menstrual cycle approximately ?

(1) 14th day
(2) 10th day
(3) 28th day
(4) 23th day

17) How many meiotic divisions are required to produce 300 sperms and 300 ovums respectively ?

(1) 75, 300
(2) 50, 300
(3) 300, 50
(4) 150, 150

18) Match the Column–A with Column–B :-

Column-A		**Column-B**	
(A)	Follicular phase	(i)	Release of secondary oocyte
(B)	Luteal phase	(ii)	Fully mature graafian follicle
(C)	Menstrual phase	(iii)	Formation of corpus luteum
(D)	Ovulatory phase	(iv)	Breakdown of endometrium

(1) A–(i), B–(ii), C–(iv), D–(iii)
(2) A–(ii), B–(iii), C–(i), D–(iv)

(3) A–(ii), B–(iii), C–(iv), D–(i)

(4) A–(iv), B–(ii), C–(i), D–(iii)

19) Which of the following is incorrectly paired with its function ?

(1) Ovary → Synthesis and secretion of steroid hormones

(2) Fimbriae → Collection of the ovum after ovulation

(3) Seminal vesicle → Produces a sugar containing fluid to nourish sperm

(4) Bartholin glands → Secrete alkaline fluid to destroys the acidity of the urethra

20) Given below are two statement : One is labelled as Assertion (A) and other is labelled as a Reason (R) : **Assertion (A) :** Blastocyst attached to the endometrium and then become embedded in the endometrium is called implantation.
Reason (R) : Trophoblast and endometrium both are adhesive in nature.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both A and R are true but R is not the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A

21) Given below are two statements : **Statement-I :** Gonadotropins increases gradually during the follicular phase of menstrual cycle.
Statement-II : Low level of progesterone are main cause of shedding endometrium during menstrual cycle.

(1) Both statement-I and statement-II are false.

(2) Statement-I is true but statement-II is false.

(3) Statement-I is false but statement-II is true.

(4) Both statement-I and statement-II are true.

22) **Assertion (A) :** Perimetrium is thin membranous layer of uterus.
Reason (R) : Perimetrium is outer layer of uterus.

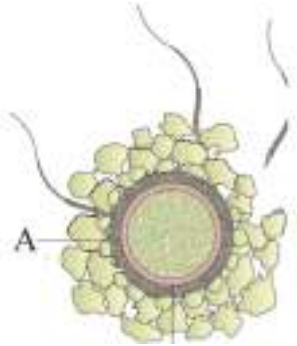
(1) Both A and R are correct and R is correct explanation of A.

(2) Both A and R are correct but R is not correct explanation of A.

(3) A is correct but R is incorrect.

(4) Both A and R are incorrect.

23) In a given diagram, identify the (A) :



(1) Plasma membrane

(2) Zona Pellucida

(3) Corona Radiata

(4) Perivitelline space

24)

Which of the following contributes in the formation of seminal plasma :-
(a) Sertoli cells
(b) Seminal vesicle
(c) Spermatogonia
(d) Leydig cells
(e) Bulbourethral gland
(f) Prostate gland

(1) only b, c, e, f

(2) only a, b, c, f

(3) only b, c, d, e, f

(4) only b, e, f

25) Read the following statements :-
(a) It is a paired structure
(b) It contains glandular tissue
(c) It has variable amount of fat
In above statement it refers to

(1) Uterus

(2) Cowper gland

(3) Breast

(4) Prostate gland

26) The sperm contains:

(1) A diploid nucleus in its head region

(2) A haploid nucleus in its middle piece

(3) A haploid nucleus in its head region

(4) Numerous mitochondria in tail region

27) First polar body is released during oogenesis:

(1) Before birth

(2) After completion of meiosis-I

(3) At the time of fertilisation

(4) At the time of formation of haploid ovum

28)

Select the hormones group which produced only during pregnancy :

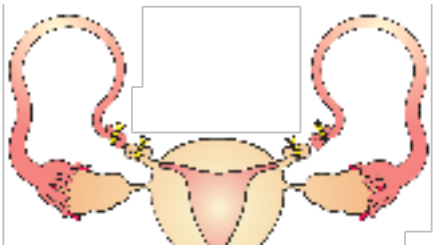
(1) Prolactin, HPL, HCG
(2) HPL, HCG, Relaxin
(3) Inhibin, Progesterone, HPL
(4) Thyroxine, Cortisol, HPL

29)

The foetal ejection reflex by the fully developed foetus and placenta triggers the release of :

(1) Progesterone from maternal ovary
(2) Relaxin from maternal corpus luteum
(3) hCG from maternal part of placenta
(4) Oxytocin from maternal pituitary

30) Identify the procedure shown in the figure given below:



(1) Tubectomy
(2) Vasectomy
(3) Ovarian cancer
(4) Uterine cancer

31) Identify the contraceptive device shown below as well as the related right place of its implantation into a woman and select the correct option together :



	Contraceptive device	**Site of implant**
(1)	LNG – 20	Fallopian tube
(2)	Lippes loop	Uterine wall
(3)	Copper–T	Uterine cavity
(4)	Multiload 375	Oviduct

(1) 1

(2) 2

(3) 3

(4) 4

32) Which of the following are natural contraceptive methods?
(A) Periodic abstinence
(B) Coitus interruptus
(C) Lactational amenorrhea
(D) M.T.P.

(1) A, B and D

(2) A, C, D

(3) A, B, C

(4) B, D

33) Which may be a complication problem of STD, when it is not timely detected ?

(1) PID

(2) Still Birth

(3) Infertility

(4) All of these

34) Match the following and give the answer for correct match:

Column-I		**Column-II**	
A.	ZIFT	i.	More than 8 blastomere stage transfer into the uterus
B.	IUT	ii.	Sperm is directly injected into the cytoplasm of ovum.
C.	GIFT	iii.	Zygote upto 8 blastomere could be tranfer into the fallopian tube.
D.	ICSI	iv.	Transfer of an ovum collected from ovary into the fallopian tube

(1) A-i B-ii C-iii D-iv

(2) A-iv B-iii C-ii D-i

(3) A-iv B-iii C-i D-ii

(4) A-iii B-i C-iv D-ii

35) Weekly oral pills for females are :-

(1) Mala-D

(2) Mala-N

(3) Saheli

(4) Gossypol

36) An emergency contraceptive pill is given to a female who is 2 weeks pregnant. This will lead to:

(1) Termination of pregnancy.
(2) Prevention of implantation
(3) Pregnancy is unaffected.
(4) Degeneration of corpus luteum due to negative feedback of LH.

37) Which is not a suitable method for population control?

(1) Oral pills
(2) IUDs
(3) Condoms
(4) M.T.P

38) Natural method of contraception include the following ?

(1) Coitus interruptus, periodic abstinence
(2) Lactational amenorrhea, IUD
(3) Periodic abstinence, Diaphragms
(4) All of these

39) The contraceptive method with almost no side effect is :-

(1) Periodic abstinence
(2) Implants
(3) IUDs
(4) Pills

40) Which of the following has been used for emergency contraception ?

(1) Diaphragm
(2) Daily tablets
(3) Saheli
(4) High progesterone and estrogen combination

41) Medical termination of pregnancy is safe during :

(1) upto 20 weeks of pregnancy
(2) upto 12 weeks of pregnancy
(3) during second trimester of pregnancy
(4) upto 24 weeks of pregnancy

42) After in vitro fertilization :

(1) Embryo with more than 8 cells are implanted in uterus.
(2) Embryo with more than 16 cells are not implanted in uterus.
(3) Embryo with 4-8 cells can be implanted in uterus.
(4) Embryo with cells more than 16 is implanted in fallopian tube

43)

Growth rate of population in India is:

(1) More than 2%
(2) Less than 2%
(3) More than 12%
(4) Less than 12%

44) Which of the following statement is **not correct** :-

(1) Progestogens alone or in combination with estrogen can also be used by females.
(2) Saheli is "once a week" pill.
(3) Pills have to be taken daily for a period of 21 days.
(4) Saheli has high side effect and low contraceptive value.

45) If the vas-deferens of a man are surgically cut or blocked :

(1) Semen will be without sperms
(2) Spermatogenesis will not take place
(3) Testosterone will disappear from blood
(4) Sperms in the semen become nonmotile

46) The world population which was around **___A___** in 1900 rocketed to about **___B___** by 2000. Identify the 'A' and 'B' :-

(1) A - 2 billion, B - 4 billion
(2) A - 2 billion, B - 6 billion
(3) A - 1 billion, B - 6 billion
(4) A - 4 billion, B - 8 billion

47) 'Saheli' is a new oral contraceptive for females.
Which of the following statements about it is correct?

(1) It is a steroidal preparation taken daily.
(2) It is a non-steroidal preparation taken once a week.
(3) It is a copper-releasing implant
(4) It is an injectable hormone

48) CuT, Cu7, and Multiload 375 are examples of:

(1) Non-medicated IUDs
(2) Copper releasing IUDs
(3) Hormone releasing IUDs
(4) Oral contraceptive pills

49) Identify the contraceptive device shown below as well as the related right place of its

implantation into a woman and select the correct option for the two together : 

	Contraceptive device	Site of implant
(1)	LNG – 20	Fallopian tube
(2)	Lippes loop	Uterine wall
(3)	Implants	Subcutanceous
(4)	Multiload 375	Uterine wall

(1) 1

(2) 2

(3) 3

(4) 4

50) **Assertion (A):-** The presence or absence of hymen is not a reliable indicator of virginity.
Reason (R):- In some women, the hymen persists even after coitus and it can also be broken by a sudden fall or jolt.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

51) Consider the following statements regarding the structure of a mature pollen grain:
I: The exine is made up of sporopollenin, which is one of the most resistant organic materials known and can withstand high temperatures, strong acids, and alkalis.
II: The intine is a thin and continuous layer made up of cellulose and pectin.
III: The germ pores are regions on the exine where sporopollenin is absent and through which the pollen tube emerges.
IV: The vegetative cell is smaller, floats in the cytoplasm of the generative cell, and has a spindle-shaped nucleus.
Which of the above statements are correct?

(1) I, II, and III only

(2) I, III, and IV only

(3) II and IV only

(4) All statements are correct

52) Consider the following statements regarding megasporogenesis and female gametophyte development in angiosperms:
I: A single Megaspore Mother Cell (MMC) differentiates in the micropylar region of the nucellus.
II: Monosporic development means the functional megaspore undergoes three sequential meiotic divisions to form an 8-nucleate embryo sac.
III: The mature 8-nucleate embryo sac in a typical angiosperm is 7-celled, containing 3 antipodals at

the chalazal end, 1 central cell with 2 polar nuclei, and an egg apparatus (1 egg cell + 2 synergids) at the micropylar end.
Which of the statements is/are correct?

(1) I only
(2) I and III only
(3) II and III only
(4) I, II and III

53)

Consider the following statements regarding outbreeding devices and pollination:
Statement I: Geitonogamy is functionally cross-pollination involving a pollinating agent, but genetically it is similar to autogamy since pollen grains come from the same plant.
Statement II: Dioecy prevents both autogamy and geitonogamy in plants.
Statement III: Self-incompatibility is a genetic mechanism that prevents self-pollen from fertilizing the ovules by inhibiting pollen germination or pollen tube growth in the pistil.
Which of the following is correct?

(1) Statement I and II only
(2) Statement II and III only
(3) Statement I and III only
(4) Statements I, II and III

54) **Assertion (A):** In castor and maize plants, autogamy is prevented, but geitonogamy can still occur.
Reason (R): Castor and maize are monoecious plants bearing unisexual male and female flowers on the same plant.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

55) **Assertion (A):** The embryo development precedes endosperm development in angiosperm seeds.
Reason (R): The primary endosperm cell undergoes repeated divisions to form a triploid endosperm tissue that provides nourishment to the developing embryo.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

56) **Assertion (A):** Apomictic seeds produce genetically variable offspring that show hybrid vigor across generations.
Reason (R): Apomixis is a form of asexual reproduction that mimics sexual reproduction, where

seeds are formed without fertilization.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(3) (A) is false, but (R) is true.

(4) (A) is true, but (R) is false.

57) **Assertion (A):** Tapetum cells are generally multinucleate and possess dense cytoplasm.
Reason (R): Tapetal cells undergo karyokinesis without being followed by cytokinesis (endomitosis/polyploidy).

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

58)

Match Column-I with Column-II regarding post-fertilization changes:

Column-I (Pre-fertilization)	**Column-II (Post-fertilization)**
(P) Ovary wall	(1) Perisperm
(Q) Persistent Nucellus	(2) Pericarp
(R) Ovule	(3) Seed coat
(S) Integuments	(4) Seed

Choose the correct option:

(1) P-2, Q-1, R-4, S-3

(2) P-1, Q-2, R-3, S-4

(3) P-2, Q-3, R-4, S-1

(4) P-4, Q-1, R-2, S-3

59)

Which of the following combinations of floral features are characteristic of **Anemophilous** (wind-pollinated) flowers?

1. Pollen grains are light, non-sticky, and produced in large numbers.
2. Well-exposed stamens and large, feathery stigmas.
3. Brightly colored petals with rich nectar glands and scent.
4. Flowers single or clustered in an inflorescence, often with a single ovule in each ovary.

(1) 1, 2 and 3 only

(2) 1, 2 and 4 only

(3) 2, 3 and 4 only

(4) 1 and 4 only

60) Which of the following structures in an unfertilized mature angiosperm embryo sac are **haploid**

(n)?

1. Synergids
2. Polar nuclei
3. Central cell
4. Functional megaspore
5. Primary Endosperm Nucleus (PEN)

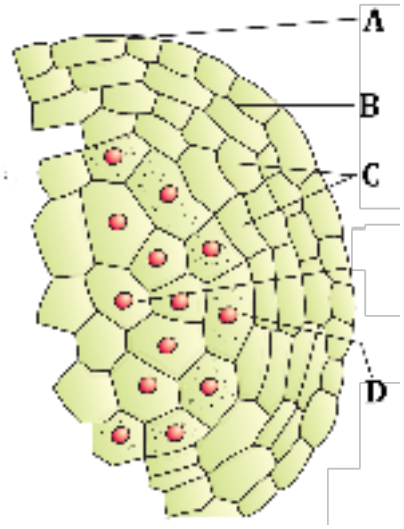
(1) 1, 2 and 4 only

(2) 1, 2, 3 and 4 only

(3) 1, 2 and 5 only

(4) All of the above

61) Refer to the simplified cross-section schematic of a young microsporangium below:



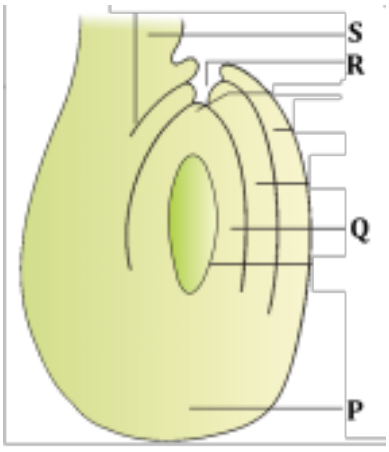
Identify layers **A, B, C, and D** in sequence from outer to inner wall layers:

(1) A = Epidermis, B = Endothecium, C = Middle layers, D = Tapetum

(2) A = Endothecium, B = Epidermis, C = Tapetum, D = Middle layers

(3) A = Epidermis, B = Middle layers, C = Endothecium, D = Tapetum

(4) A = Tapetum, B = Middle layers, C = Endothecium, D = Epidermis



62) Analyze the structural representation of an **Anatropous Ovule**: Which of the following correctly pairs the label with its description?

(1) **P** = Micropylar pole; represents basal part of ovule

(2) **Q** = Nucellus; central mass of parenchymatous tissue with abundant food reserves

(3) **R** = Chalazal pole; small opening where integuments encircle completely

(4) **S** = Hilum; stalk connecting ovule to placenta

63) If the chromosome number in the root tip cells of a diploid angiosperm plant (2n) is **24**, what will be the chromosome number in its **aleurone layer**, **synergid cell**, and **integument cell** respectively?

(1) 36, 12, 24

(2) 24, 12, 36

(3) 36, 24, 12

(4) 12, 24, 36

64) Which of the following is **TRUE** regarding aquatic plants and pollination?

(1) All aquatic plants use water for pollination.

(2) In *Vallisneria*, female flowers remain submerged, and pollen is released inside water.

(3) In *Water Hyacinth* and *Water Lily*, flowers emerge above water and are pollinated by insects or wind.

(4) In *Zostera* (seagrass), female flowers float on the surface to reach dry pollen.

65) The phenomenon in *Citrus* and mango where additional embryos develop from diploid nucellar cells protruding into the embryo sac is known as:

(1) Parthenocarpy

(2) Adventive polyembryony

(3) Apospory

(4) Parthenogenesis

66) During pollen-pistil interaction, the filiform apparatus of synergids plays a crucial role by:

(1) Secreting enzymes to digest the stigma surface.

(2) Guiding the entry of the pollen tube into the synergid.

(3) Preventing polyspermy during double fertilization.

(4) Fusing with polar nuclei to form endosperm.

67) What is the function of **endothecium** in a mature fibrous anther?

(1) Secretion of sporopollenin

(2) Hygroscopic nature aiding in dehiscence along the stomium

(3) Nourishment of microspore tetrads

(4) Synthesis of pollen exine

68) An angiospermic plant has 18 chromosomes in its microspore mother cell. How many chromosomes will be present in its **endosperm cell** and **generative cell**?

(1) 27 and 9

(2) 18 and 9

(3) 36 and 18

(4) 27 and 18

69) Which of the following statements about **pollen bank** storage conditions is correct?

(1) Pollen grains are stored in liquid nitrogen at –196°C.

(2) Pollen grains are stored in dry ice at –80°C.

(3) Pollen grains are stored in liquid helium at –269°C.

(4) Pollen grains are stored in formalin solution at 0°C.

70) The primary endosperm nucleus (PEN) in typical angiosperms undergoes free-nuclear divisions in early stages. An example of **free-nuclear endosperm** development followed by cellularization is:

(1) Coconut water (free nuclear) and surrounding white kernel (cellular)

(2) Pea seed

(3) Groundnut seed

(4) Castor seed pericarp

71) What is the ploidy level of the cells of integument, MMC, microspore and Male Gametophyte respectively?

(1) 2n, 2n, n, n

(2) 2n, n, n, 2n

(3) n, 2n, n, 2n

(4) 2n, 2n, 2n, n

72) Chasmogamous flowers are those which:

(1) Never open at all

(2) Open with exposed anthers and stigma

(3) Produce seeds without pollination

(4) Are strictly self-pollinated

73) Parthenocarpic fruits are developed without fertilization and are seedless. Which plant growth regulator is commonly applied to induce parthenocarpy synthetically?

(1) Abscisic acid

(2) Auxins and Gibberellins

(3) Ethylene gas

(4) Cytokinins only

74) Which part of the seed serves as the persistent remnant of the **nucellus** in seeds like black pepper and beet?

(1) Scutellum

(2) Perisperm

(3) Pericarp

(4) Tegmen

75) In monocot seeds (e.g., Grass family), the single shield-shaped cotyledon attached to the embryonal axis is called:

(1) Coleoptile

(2) Scutellum

(3) Coleorhiza

(4) Epiblast

76) In grass embryos, the radical and root cap are enclosed in an undifferentiated sheath called **(X)**, and the shoot apex and leaf primordia are enclosed in a foliar sheath called **(Y)**.

(1) X = Coleoptile, Y = Coleorhiza

(2) X = Coleorhiza, Y = Coleoptile

(3) X = Epiblast, Y = Scutellum

(4) X = Hypocotyl, Y = Epicotyl

77) What is the major advantage of **apomictic seeds** in agriculture?

(1) They require manual cross-pollination every season.

(2) Farmers do not need to buy costly hybrid seeds every year because hybrid traits do not segregate in apomictic offspring.

(3) They increase genetic recombination and diversity in offspring.

(4) They produce small, infertile seeds easy for processing.

78) In 60% of angiosperms, pollen grains are shed at the **2-celled stage**. What are these two cells?

(1) Two male gametes

(2) One vegetative cell and one generative cell

(3) One generative cell and one tube nucleus

(4) Two vegetative cells

79) What is the fate of the **antipodal cells** inside the embryo sac after double fertilization?

(1) They turn into endosperm

(2) They degenerate

(3) They develop into secondary embryos

(4) They fuse with male gametes

80) Which of the following sets contains **only non-albuminous seeds**?

(1) Pea, Groundnut, Beans

(2) Castor, Sunflower, Maize

(3) Wheat, Barley, Castor

(4) Cotton, Maize, Castor

81) Emasculation is the process of removing anthers from the flower bud before dehiscence. Which type of flower **does NOT** require emasculation during artificial hybridization?

(1) Bisexual flowers

(2) Monoclinous flowers

(3) Unisexual female flowers

(4) Hermaphrodite flowers

82) False fruits develop from floral parts other than or in addition to the ovary. Which of the following is an example of a **false fruit** where the **thalamus** contributes to fruit formation?

(1) Mango

(2) Apple and Strawberry

(3) Cucumber

(4) Banana

83) Viability of pollen grains varies greatly. Pollen grains of cereals like **wheat and rice** lose viability within **(X)**, while those of **Rosaceae, Leguminosae, and Solanaceae** retain viability for **(Y)**.

(1) X = 30 minutes, Y = several months

(2) X = 24 hours, Y = 30 minutes

(3) X = several months, Y = 30 minutes

(4) X = 10 minutes, Y = 2 days

84) During double fertilization in angiosperms:

(1) One male gamete fuses with the egg (Syngamy), and the other male gamete fuses with two polar nuclei (Triple Fusion).

(2) Both male gametes fuse with two egg cells simultaneously.

(3) One male gamete fuses with the synergid, and the other fuses with antipodals.

(4) One male gamete fuses with the central cell membrane, and the other degenerates.

85)

In an artificial hybridization experiment on a bisexual flower, arrange the steps in the correct chronological order:

1. Bagging the flower bud.
2. Dusting desired pollen grains on the receptive stigma.
3. Emasculation of the anthers from the flower bud.
4. Re-bagging the flower and allowing fruit development.

(1) 3 → 1 → 2 → 4

(2) 1 → 3 → 2 → 4

(3) 3 → 2 → 1 → 4

(4) 2 → 3 → 1 → 4

86) The hilum is a scar on the :

(1) Fruit, where it was attached to pedicel

(2) Fruit, where style was present

(3) Seed, where micropyle was present

(4) Seed, where funicle was attached

87) Match column-I with column-II and select the correct option using the codes given below :

Column-I		**Column-II**	
a	Pistils fused together	i	Gametogenesis
b	Formation of gametes	ii	Pistillate
c	Typical anther	iii	Syncarpous
c	Unisexual female flower	iv	Bilobed

	a	**b**	**c**	**d**
(1)	i	ii	iv	iii
(2)	iii	i	iv	ii
(3)	iv	iii	i	ii
(4)	ii	i	iv	iii

(1) 1

(2) 2

(3) 3

(4) 4



88) Identify the correct description about the given figure :

(1) Wind pollinated plant inflorescence showing flowers with well exposed stamens.

(2) Water pollinated flowers showing stamens with mucilaginous covering.

(3) Cleistogamous flowers showing autogamy

(4) Compact inflorescence showing complete autogamy.

89) Identify the correct set of statements with respect to angiosperms:-
(a) A microsporophyll contains stigma, style and ovary
(b) All megaspores produced from megaspore mother cell take part in embryo sac formation.
(c) Hilum is the place where ovule joins with funicle
(d) Basal part of an ovule is represented as micropylar end
(e) Perisperm is present in black pepper.
Choose the correct answer from the options given below

(1) only c and e are correct

(2) a, b, c and e are correct

(3) c, d and e are correct

(4) only c is correct

90) The arrangement of the nuclei in a normal embryosac in the dicot plants is :-

(1) 2 + 4 + 2

(2) 3 + 2 + 3

(3) 2 + 3 + 3

(4) 3 + 3 + 2

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	4	3	4	3	3	4	3	2	1	1	2	4	1	4	2	4	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	3	1	3	2	1	4	4	2	2	4	4	4	2	4	1	2	2	2
Q.	41	42	43	44	45															
A.	2	2	3	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	1	1	4	4	2	4	2	4	3	3	3	2	3	2	1	2	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	3	2	2	2	4	4	2	3	1	1	1	2	2	2	4	3	3	1
Q.	86	87	88	89	90															
A.	1	2	1	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	3	4	2	1	2	2	2	4	4	3	2	3	3	4	1	3	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	2	4	3	3	2	2	4	1	3	3	4	4	3	3	4	1	1	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	2	4	1	2	2	2	3	1	1	2	4	1	4	3	1	1	2	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	2	1	3	2	2	2	1	1	1	1	2	2	2	2	2	2	2	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	1	1	1	4	2	1	1	2										

SOLUTIONS

PHYSICS

1) **Explain Question:**
Question ask to find ratio of tension in string

Concept:
Motion of bodies connected with string.

Solution:

$$a = \frac{F}{3m}$$

$$T_{PQ} = 2m \times \frac{F}{3m}$$

$$= \frac{2F}{3}$$

$$T_{QR} = m.\frac{F}{3m}$$

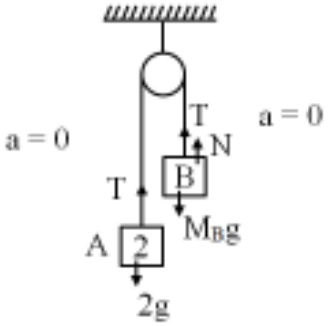
$$= \frac{F}{3}$$

$$\frac{T_{PQ}}{T_{QR}} = \frac{2F}{3} \times \frac{3}{F}$$

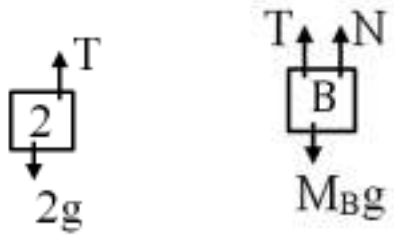
= 2 : 1

Final Answer: option(2)

2) **Explain the question :**



We have to calculate mass of block B(M_B) = ?
Concept : This question is based on motion of bodies connected with pulley system without spring.
Solution :



$T = 2g$ $T + N = M_B g$

$2g + 10 = M_B \times (10)$

$20 + 10 = M_B \times 10$

$M_B = \frac{30}{10} = 3kg$

Final answer : (3)

3) **Question Explanation:**
This problem asks to identify the correct statement(s) regarding the motion of on object based on dot products involving $(\vec{a})$, velocity $(\vec{v})$ and position unit vector $(\vec{r})$.

Concept:
This question is based on kinematics, vector.

Solution:
(A) $\hat{a} \cdot \hat{v} = 1$ means angle between $\hat{a}\&\hat{v}$ is zero hence speed will increase.
(B) $\hat{a} \cdot \hat{v} = -1$ means angle between $\hat{a}\&\hat{v}$ is 180° hence speed will decrease.
(C) $\vec{r} \cdot \vec{v} = -1$ means angle between $\hat{a}\&\hat{v}$ is 180° hence object will move towards origin.

Final Answer:
The correct option is (4)

4) **Question Explanation:** Find the velocity after t seconds if a particle moving with uniform acceleration travels a sum of 100 cm in the tth and $(t+1)^{th}$ seconds.
Concept Involved: Kinematic equations of motion for uniform acceleration (v = u + at and distances in specific time intervals).
Solution:
a = f (given), Distance in the nth second is

$$S_n = u + \frac{f}{2}(2n-1)$$

$$S_t = u + \frac{f}{2}(2t-1)$$

$$S_{(t+1)^{th}} = u + \frac{f}{2}[2(t+1)-1]$$

$$= u + \frac{f}{2}(2t+1)$$

Sum of distances

$S_t + S_{(t+1)} = 100$

$$\left[u + \frac{f}{2}(2t-1)\right] + \left[u + \frac{f}{2}(2t+1)\right] = 100$$

$2u + 2ft = 100$

$u + ft = 50$

Using 1st of motion $v = u + ft$

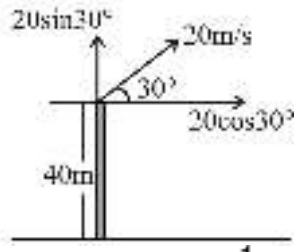
$v = 50$ cm/s

Final Answer: (3).

5)

Explain the question : To find the time after which the projected body will reach the ground.
Concept : This question is based on projection from height.

Solution :-



$h = -(u \sin\theta)\, t + \frac{1}{2} g t^2$

$40 = -(20 \sin 30°)\, t + \frac{1}{2} \times 10 \times t^2$

t = 4 sec

Final answer :- (4)

6) **Asking About :**

The scenario where the normal reaction between the stacked blocks is minimized.

Concept :

The normal force between the blocks can be calculated by newtons second law

Formula :

N = m(g + a)

• Use '+' if acceleration is upwards (increasing weight).

• Use '–' if acceleration is downwards (decreasing weight).

Calculation :

• Case A : N = 2 × 10 = 20 N

• Case B : 2 × (10 + 2) = 24 N

• Case C : N = 2 × (10 – 3) = 14 N (Minimum)

• Case D : Normal force due to horizontal force is higher than Case C

Final Answer: (3)

7) **Explaining:** Find the percentage error in density

Concept: Error propagation

Solution:

$$\text{Density} = \rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3}$$

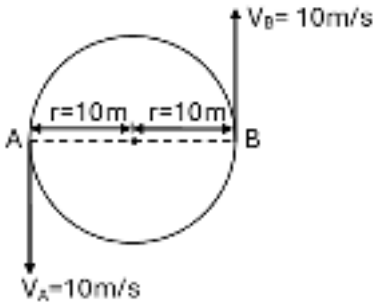
$$\Rightarrow \frac{\Delta\rho}{\rho} \times 100\% = \left[\frac{\Delta M}{M} \times 100\% + 3\frac{\Delta R}{R} \times 100\%\right]$$

$$= [2\% + 3 \times 1\%] = 5\%$$

Final Answer: (3)

8)

Question Explanation :



In this question, for journey from A to B, we have to find $|\vec{a}_{avg}|$

Concept :

This question is based on Average acceleration.

Solution :

$$\vec{a}_{avg} = \frac{\Delta\vec{v}}{\Delta t}$$

$$= \frac{\vec{v}_B - \vec{v}_A}{\left(\frac{\pi r}{v}\right)} \quad \left(\Delta t = \frac{\pi r}{v}\right)$$

$$= \frac{10(+\hat{j}) - 10(-\hat{j})}{\left(\frac{\pi \times 10}{10}\right)} = \frac{10(\hat{j}) + 10(\hat{j})}{\pi}$$

$$= \frac{20(\hat{j})}{\pi} m/s^2$$

$$|\vec{a}_{avg}| = \frac{20}{\pi}\ m/s^2$$

Final Answer : Option (4)

9)

Explain Question: find the average impulse imparted to a bullet. The force acting on the bullet is a function of time, and the force drops to zero when the bullet leaves the gun barrel.

Concept: This question is based on impulse

Formula : $J = \int_0^{t_f} F(t)dt$

Solution: Time when the force becomes zero

$0 = 600 - 2 \times 10^5 t_f$

$2 \times 10^5 t_f = 600$

$$t_f = \frac{600}{2 \times 10^5} = 300 \times 10^{-5} = 3 \times 10^{-3}\ s$$

$$\text{Impulse} = \int_0^{t_f} Fdt$$

$$= \int_0^{t_f} (600 - 2 \times 10^5)dt$$

$$= \left[600t - 2 \times 10^5 \frac{t^2}{2}\right]_0^{t_f=0.003}$$

$= [600 \times 0.003 - 10^5 (0.003)^2]$

$= 1.8 - 0.9 = 0.9$ N-s

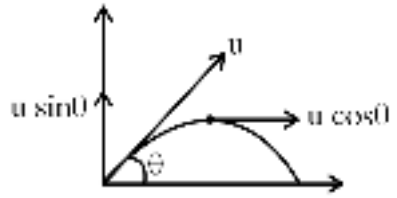
Final Answer : (3)

10)

Explain the Question : In this question we have to find the time of flight of the projectile whose initial speed is 'u' and at the highest point of its velocity is $\frac{\sqrt{3}}{2}u$

Concept : This question is based on ground to ground projectile motion.

Solution :



At the highest point, velocity is ucosθ

$\Rightarrow u\cos\theta = \frac{\sqrt{3}}{2}u$ (as given in the question)

$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30°$

Time of flight of projectile, $T = \frac{2u\sin\theta}{g}$

$\Rightarrow T = \frac{2u\sin 30°}{g} \Rightarrow \frac{2u \times \frac{1}{2}}{g} = \frac{u}{g}$

Final answer : 2

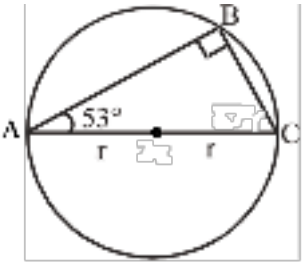
11)

Explain question : To find the distance covered by the particle, when it travels A to B and then B to C.

Concept : This question is based on distance and displacement.

Solution :

In ΔABC



$AB = 2r\cos 53° = 2\left(\frac{3}{5}\right)r$

$BC = 2r\sin 53°$

$= 2r\left(\frac{4}{5}\right)$

AB + BC

$= \frac{6}{5}r + \frac{8}{5}r$

$= \frac{14r}{5}$

Final answer : 1

12) **Explaining:** Find the reading of screw gauge
Concept: Screw gauge
Solution:
Reading of screw guage
$= 3 \text{ mm} + 25 \times 0.01 \text{ mm} \left(\text{L.C.} = \frac{1}{100}\text{mm} = 0.01\,\text{mm}\right)$
$= 3.25$ mm
Final Answer: (1)

13) **Question Explanation:**
Finding the acceleration of a two-block system with friction.

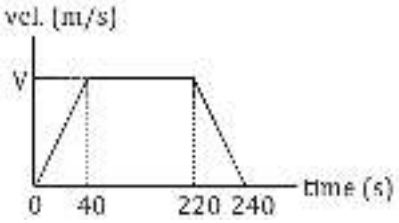
Concept:
Newton's Second Law and Kinetic Friction.

Solution:
$f = \mu N = \mu(2Mg)$
$F_{pulling} = Mg$
$F_{net} = F_{pulling} - f$
$(M + 2M)a = Mg - 2\mu Mg$
$3Ma = Mg(1 - 2\mu)$
$a = \frac{g(1-2\mu)}{3}$

Final Answer: Option(2)

14)

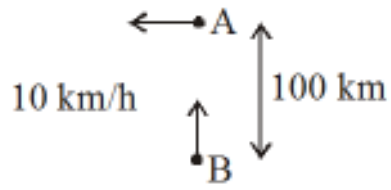
Explaining Question : Asking about the max speed of the train travelling between two stations with different accelerations during journey
Concept: This question is based on equation of motion.
Solution :



Area of v – t graph = displacement
$2250 = \frac{1}{2}(240 + 180) \times V$
$V = 10.71$ m/s
Final Answer : Option (4)

15) **Explaining:** Find the time after which the distance between A & B becomes shortest
Concept: Relative motion
Solution:

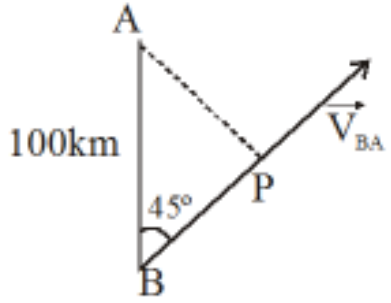
$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$



$= 10\hat{j} - (-10\hat{i})$

$= 10\hat{j} + 10\hat{i}$

$V_{BA} = 10\sqrt{2}$ km/h



Time = $\frac{\text{distance}}{\text{speed}}$

$= \frac{50\sqrt{2}}{10\sqrt{2}} = 5$h

Final Answer: (1)

16)

Explain Question : We need to find the wrong statements (s) about the motion of the ball and the net force acting on it.

Concept : This question is based on Newtons second law (F = ma)

Solution :

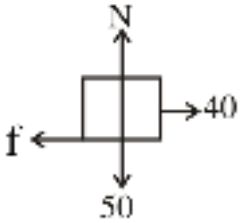
$\vec{F} = m\vec{a} = m\vec{g}$

Force of gravity acts always vertically downward.
The net force and the net acceleration are downward, throughout the motion of the ball.

Final Answer : option (4)

17)

Explain Question : A 5 kg box is subjected to a force of 40 N. The static and kinetic coefficients of friction are provided. The question asks for the acceleration, normal force, limiting friction, and kinetic friction of the box.
Concept : This question is based on Newton's Second Law of Motion, frictional forces.
Solution :



N = 50 newton

$f_L = 0.6 \times 50 = 30$ N

$f_k = 0.5 \times 50 = 25$ N

$a = \frac{40-25}{5} = 3\ m/s^2$

Final answer : Option (2)

18) **Explaining:** Find the projection on y-axis

Concept: Application of dot product

Solution:

Projection = $\frac{(3\hat{i}+4\hat{k}).\hat{j}}{|\hat{j}|} = 0$

Final Answer: (4)

19) **Question Explanation:**

Calculates overtaking time using relative motion.

Concept:

Relative Velocity and Total Distance.

Solution:

When a faster train overtakes a slower one, the total distance to be covered is the sum of both lengths. The effective speed is the difference between their velocities because they are moving in the same direction.

Calculation:

Relative velocity of overtaking = 40 ms^{-1} – 30 ms^{-1}

= 10 ms^{-1}. Total distance covered with this relative velocity during overtaking will be

= 100 m + 200 m = 300 m.

$\text{Time taken} = \frac{300m}{10ms^{-1}} = 30\,s.$

Final Answer:

Option (1)

20) **Question Explanation :** In this question we have to find the maximum area covered by the bullets fired in all possible directions.

Concept: This Question is based on ground to ground projectile.

Solution:

Area is maximum when range is maximum

Area = $\pi(R_{max})^2$

$R_{max} = \frac{u^2 \sin 2\theta}{g}$ For R_{max}, $\theta = 45°$

$R_{max} = \frac{u^2}{g}$

$A_{max} = \pi\left(\frac{u^2}{g}\right)^2$

Final Answer: Option (4)

21)

Question Explanation:
The question asks to find the time for which velocity is uniform in the given (v-t) graph.

Concept:
Graphical analysis of (v-t) graph.

Solution: Slope of v-t graph gives the acceleration
Acceleration a = Slope = tan θ = tan 45º = 1 m/s^2
From starting point to A
By the equation
$v = u + at$
$3 = 0 + 1 \times t$
$t = 3 \text{ sec}$
Similarly $t_{BC} = 3 \text{ sec}$

The Motion is uniform from A to B and overall time of journey is 10 sec.

Hence,
$t_{AB} = 10 - (3 + 3) = 4 \text{ sec}$

Final Answer: Option (2)

22) **Explaining:** Find value of $\left(\vec{A} + \vec{B}\right) . \left(\vec{A} \times \vec{B}\right)$
Concept: Dot and cross product
Solution:
Let $\vec{R} = \vec{A} + \vec{B}$ and $\vec{C} = \vec{A} \times \vec{B}$
thus, $\vec{C}$ is $\perp$ to plane of $\vec{A}$ and $\vec{B}$ and $\vec{C} \perp \vec{R}$ also
$\vec{C}.\vec{R} = 0 \Rightarrow \left(\vec{A} + \vec{B}\right) . \left(\vec{A} \times \vec{B}\right) = 0$
Final Answer: (1)

23)

Explain Question :
Given the equation for the path of a projectile, find its horizontal range.
Concept:
The horizontal range (R) of a projectile is the value of x when the vertical displacement (y) returns to zero (y = 0).
Solution:
Equation: $y = 12x - \frac{3}{4}x^2$
Set y = 0: $0 = 12x - \frac{3}{4}x^2$

Factor out x: $x\left(12-\frac{3}{4}x\right)=0$

Solve for non-zero x: $12=\frac{3}{4}x \Rightarrow x=\frac{12\times 4}{3}=\mathbf{16}\ \text{m}$

Final Answer: Option 3

24) **Question Explanation :** Find dimensional formula of Mass in terms of E, V, and T.
Concept: This question is based on dimensional analysis
Solution:
$m \propto E^x V^y T^z$
$M^1L^0T^0 \propto [ML^2T^{-2}]^x[LT^{-1}]^y[T]^z$
$M^1L^0T^0 \propto M^x L^{2x+y} T^{-2x-y+z}$
Compare power of M $\Rightarrow x = 1$

Compare power of L

$2x + y = 0$
$2(1) + y = 0 \Rightarrow y = -2$
Compare power of T

$-2x - y + z = 0$
$-2(1) - (-2) + z = 0$
$z = 0$
$[M]=[E]^1[V]^{-2}[T]^0$

Final Answer: Option (1)

25)

Explain Question:
Question ask to find acceleration of boy.

Concept:
Motion of bodies connected with pulley system.

Solution:
Tension on string, $T = 100\ g \sin 53°$

$= 100 \times 10 \times \frac{4}{5}$

$= 800\ N$

So $T - 50g = 50\ a$

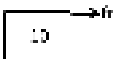
$800 - 500 = 50\ a$

$\Rightarrow a=\frac{300}{50}\text{m/s}^2 \Rightarrow a = 6\ \text{m/s}^2$

Final Answer: Option(3)

26) **Explain Question:**
We have to calculate frictional force between the blocks.
Concept:
Two block system in friction.
Solution:

Common acceleration $a_c = \frac{10}{10+5} = \frac{2}{3}$



$fr = m_2 a_C = 10 \times \frac{2}{3} = \frac{20}{3}N$

$f_L = \mu_s N = 0.5 \times 5 \times 10 = 25N$

$f_L > f_r$

Both blocks moves together

$f_r = \frac{20}{3}N$

Final Answer: Option (2)

27)

Explain question : evaluate two statements—an **Assertion** and a **Reason**—related to the forces acting on a body in contact with a surface.

Concept : This question is based on contact force

Explanation :

$F_C = \sqrt{N^2 + f^2}$

Assertion (A) : Since the normal reaction force (N) is always non-zero for a body in contact with a surface, Therefore, contact force will always be greater than friction force

Reason (R): This is the definition of contact force. It is the vector sum (or resultant) of the normal reaction and friction force.

Final answer :- (1)

28) **Explaining:** Find the time taken by object to again hit the floor

Concept: Motion under gravity

Solution:

$g_{eff.} = g + a$

$T = \frac{2u}{g_{eff.}} = \frac{2 \times 18}{12} = 3\,sec$

Final Answer: (4)

29) **Explaining the Question:** This question requires matching velocity-time equations from List-I with their corresponding acceleration or position values from List-II at t = 1s.

Concept: This question is based on application of calculus

Solution :

Case (I) : $V = \frac{t^2}{2} + 1$

Position at t = 1 sec, $x(t) = \int_0^1 \left(\frac{t^2}{2} + 1\right) \cdot dt = \left[\frac{t^3}{6} + t\right]_0^1 = \frac{1}{6} + 1 = \frac{7}{6}$

I → S

Case (II):

$V = 3t^3$

$x(t) = \int_0^1 3t^3 dt = \left[\frac{3t^4}{4}\right]_0^1 = \frac{3}{4}$

II → R

Case (III):
V = 2t

$$x(t) = \int_0^1 2t\,dt = \left[t^2\right]_0^1 = 1$$

III → U
Case (IV):
V = 2t + 2

$$x(t) = \int_0^1 (2t+2)dt = \left[t^2 + 2t\right]_0^1 = 1 + 2 = 3$$

IV → T
Therefore, the correct combination is:-
I-S;II-R;III-U;IV-T
Final Answer : (4)

30) **Explaining:** Select correct v-t graph for given x-t graph
Concept: x-t graph analysis
Solution:
At t = 0 ; curve goes upwards, slope → positive
From t = 0 to t = T ; positive slope is decreasing, velocity is decreasing
At t = T ; slope is zero, velocity is zero (v = 0)
From t = T to t = 2T ; slope becomes negative and increasingly steep, thus velocity is negative and continues to decrease.
Final Answer: (2)

31) **Question Explanation:**
We have to calculate values of x and y.

Concept:
Cross product.

Solution:
For parallel vectors, compare the rations of corresponding components

$$\frac{6}{5} = \frac{x}{-6} = \frac{-2}{-y}$$

by solving this

$$x = \frac{-36}{5},\ y = \frac{5}{3}$$

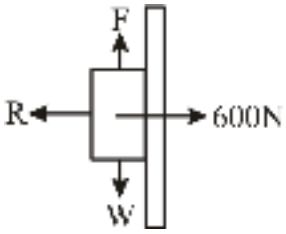
Final Answer: Option(2)

32) **Explaining :** Find the force needed so that rope become horizontal
Concept : This question is based on To make string horizontal, there must be no net force in vertical direction
Solution :
To make string horizontal, there must be no net force in vertical direction. As, there is no force to balance weight, so rope can not become horizontal.

Final Answer : (4)

33) **Explaining:** Find the acceleration of man to sliding down
Concept: Motion against vertical wall
Solution:
Friction = μR = 0.5 × 600 = 300 N
Weight = 600 N

$$a = \frac{F-W}{M}$$



$$= \frac{600-300}{60}$$

$$= \frac{300}{60} = 5\ m/s^2$$

Final Answer: (4)

34) **Explaining:** Find average speed of particle
Concept: Average speed
Solution:

$$V_{av} = \frac{V_1t_1 + V_2t_2 + V_3t_3}{t_1 + t_2 + t_3}$$

$$\Rightarrow \quad 2.5V = \frac{Vt_0 + (2V)t_0 + (3V)T}{t_0 + t_0 + T}$$

$$\Rightarrow \quad 5t_0 + 2.5\ T = 3t_0 + 3T$$

$$\Rightarrow \quad 2t_0 = 0.5\ T$$

$$\Rightarrow \quad T = \frac{2t_0}{0.5} = 4t^0$$

Final Answer: (4)

35) **Explaining:** Find the percentage error in time period
Concept: Error propagation
Solution:
Given, ΔT = 0.1 s
T = 20 s

$$\frac{\Delta T}{T} \times 100\%$$

$$= \frac{0.1}{20} \times 100$$

= 0.5 %
Final Answer: (2)

36)

Question Explanation:

The question asks for the physical quantity whose dimensions are the same as $\frac{A^3}{B}$ given the position function $x(t) = \frac{A}{B}\left(1 - e^{-At}\right)$.

Concept:

Dimensional Analysis

The argument of an exponential function must be dimensionless, which helps determine the dimensions of constants.

Solution:

Given:

$x(t) = \frac{A}{B}\left(1 - e^{-At}\right)$

Since x represents position, its dimension is length [L].

The exponent –A t must be dimensionless:

$[A][t] = 1 \Rightarrow [A] = T^{-1}$

Now,

$x = \frac{A}{B}$

So,

$\frac{A}{B} = L$

$[B] = \frac{A}{L} = \frac{T^{-1}}{L}$

Now find dimensions of $\frac{A^3}{B}$

$\frac{\left(T^{-1}\right)^3}{T^{-1}/L} = \frac{T^{-3}}{T^{-1}} \times L = LT^{-2}$

This is the dimension of acceleration.

Final Answer:

Option (4)

37) **Explaining:** We have to find area of ellipse

Concept: Coordinate geometry

Solution:

Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{x^2}{9} + \frac{y^2}{4} = \frac{x^2}{3^2} + \frac{y^2}{2^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

a = 3 ; b = 2

π ab = π × 3 × 2 = 6π

Final Answer: (1)

38) **Question Explanation:** In this question, we are asked to find the unit of kinetic energy in the given system of units.

Concept: Conversion between systems of units (Dimensional analysis)

Solution :

Dimensional Formula of Kinetic Energy (K E) :

$[KE] = \left[M^1L^2T^{-2}\right]$

Unit Conversion Formula:

$$n_2 = n_1 \left[\frac{M_1}{M_2}\right]^1 \left[\frac{L_1}{L_2}\right]^2 \left[\frac{T_1}{T_2}\right]^{-2}$$

Given Values:

- New System (n_1 = 1 unit): M_1 = 100 g, L_1 = 1 cm, T_1 = 10s
- CGS System (n_2 in erg): M_2 = 1 g, L_2 = 1 cm, T_2 = 1 s.

Calculation:

$$n_2 = 1 \times \left(\frac{100\,g}{1\,g}\right)^1 \times \left(\frac{1\,cm}{1\,cm}\right)^2 \times \left(\frac{10\,s}{1\,s}\right)^{-2}$$

$$n_2 = 1 \times 100 \times (1)^2 \times \frac{1}{10^2}$$

$$n_2 = 100 \times \frac{1}{100} = 1erg$$

Final answer: (2)

39) **Explaining:** Find average velocity of car

Concept: Average velocity (v-t graph)

Solution:

Average velocity = $\frac{\int_0^{25} v\,dt}{25-0} = \frac{\text{Area of v-t graph between t=0 to t = 25 s}}{25}$

= $\frac{1}{25}\left[\left(\frac{25+10}{2}\right)(20)\right] = 14\ m/s$

Final Answer: (2)

40) **Explaining :** Find α_{max}

Concept : This question is based on tension force

Solution :

$T_C = M_1g + M_2(g+a)$

$260 = 10 \times 10 + 10\,(10 + \alpha)$

$\Rightarrow \alpha = 6\ m/s^2$

Final Answer : (2)

41) **Question Explanation :** Calculate the **fictitious (pseudo) force** acting on person B (m_B = 50 kg) as observed by person A (m_A = 70 kg) at time t = 1 s, given their respective velocities $\vec{v}_1 = 3t^2\hat{i} - \frac{4}{3}t^6\hat{j}$ and $\vec{v}_2 = 3\left(t^2 - 5t\right)\hat{i}$.

Concept: Pseudo force.

Solution:

$$\vec{a}_A = \frac{d\vec{v}_1}{dt} = 6t\,\hat{i} - 8t^5\hat{j}$$

At t = 1 sec ; $\vec{a}_A = 6\hat{i} - 8\hat{j}\ m/s^2$

Pseudo force acting on person B as observed by person A = $-m_B\vec{a}_A$

= $-50\left[6\hat{i} - 8\hat{j}\right] = 50\left[8\hat{j} - 6\hat{i}\right] = 100\left[4\hat{j} - 3\hat{i}\right]$ N

Final Answer: Option (2)

42) **Question Explaination:** The problem requires determining the diameter of a cylinder by calculating the caliper's least count from given scale dimensions, combined with main scale reading and Vernier division coincidence.

Concept Based: This question is based on Vernier Calliper.

Solution:
50 divisions = 2.45 cm

1 division = $\frac{2.45}{50} = 0.049\text{cm}$

least count = 1 MSD –1VSD
= 0.05 – 0.049
= 0.001 cm.
So vernier reading = 0.001 × 24 = 0.024 cm.
Therefore diameter of cylinder
= 5.10 + 0.024 = 5.124 cm.

Final Answer: Option(2)

43)

Explanation the Question:
Given, $x = t^3 - 75\ t$
we have to calculate force at t = 4 second.

Concept:
This question is based on Newton's second law of motion [F = ma]

Solution:
$x = t^3 - 75\ t$

$v = \frac{dx}{dt} = 3t^2 - 75$

$a = \frac{dv}{dt} = 6t$

$a_{t=4} = 6(4) = 24$
So, F = ma = 6 × 24
F = 144 newton

Final Answer: Option (3)

44) **Question Explanation**
In this question, we are asked to find the least count (LC) of the vernier calipers using the given main scale and vernier scale relation.

Concept
Least count of Vernier Calipers

Solution

Given,

$1\text{MSD} = 0.5\ \text{mm}$

Also,

$20\text{VSD} = 19\text{MSD}$

Hence,

$1\text{VSD} = \frac{19}{20}\text{MSD} = 0.95\text{MSD}$

Therefore,

$1\text{VSD} = 0.95 \times 0.5 = 0.475\ \text{mm}$

Least count,

$\text{LC} = 1\text{MSD} - 1\text{VSD} = 0.5 - 0.475 = 0.025\ \text{mm}$

Final answer : Option (2)

45) **Explaining :**

A stone is projected **horizontally** from the top of a tower with a speed of 20 m/s. It strikes the ground at a horizontal distance of 400 m. The question asks for the **time of flight** of the stone.

Concept : This question is based on horizontal projectile motion

Solution :

· Initial horizontal velocity, u_x=20 m/s.

· Horizontal distance (Range), R=400 m. Since there's no horizontal acceleration, the horizontal distance is given by: $R=u_x \times T$

where T is the time of flight. 400 m=20 m/s × $T = \frac{400\text{m}}{20\text{m/s}} = 20\text{s}$

Final Answer : (2)

CHEMISTRY

46)

Explaining : In a reversible reaction, two substances are in equilibrium. If the concentration of each one is doubled, the equilibrium constant will be

Concept : This question is based on application of equilibrium constant.

Solution : Value of K_{eq} depends only on Temperature.

Final Answer : (4)

47)

Explaining : The equilibrium constant for the reaction, $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 4×10^{-4} at 2000 K. In presence of a catalyst, equilibrium is attained ten times faster. Therefore, the equilibrium constant, in presence of the catalyst, at 2000 K is:

Concept : This question is based on application of equilibrium constant.

Solution : K_{eq} is independent of catalyst.

Final Answer : (2)

48)

Explaining : Partial pressure of O_2 in the reaction $2Ag_2O$ (s) ⇌ 4 Ag (s) + O_2 (g) is
Concept : This question is based on Law of mass action.
Solution : $K_p = P_{O_2}$
Final Answer : (1)

49)

Explaining : A quantity of PCl_5 was heated in a 10 dm^3 vessel at 250°C : PCl_5 (g) ⇌ PCl_3 (g) + Cl_2(g). At equilibrium, the vessel contains 0.1 mole of PCl_5 and 0.2 mole of Cl_2. The equilibrium constant of the reaction is.
Concept : This question is based on Law of mass action.
Solution :

PCl_5 ⇌ PCl_3 + Cl_2
0.1 mol 0.2 0.2

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} \Rightarrow K_c = \frac{\frac{0.2}{10} \times \frac{0.2}{10}}{\frac{0.1}{10}} = 0.04$$

Final Answer : (1)

50)

Explaining : A + B ⇌ C + D. If initially the concentrations of A and B are equal but at equilibrium, concentration of D will be twice that of A, then what will be the equilibrium constant of the reaction ?
Concept : This question is based on Law of mass action.
Solution :

	A	+	B	⇌	C	+	D
t_0	a		a		–		–
t_{eq}	a–x		a–x		x		x

[D] = 2 × [A] ⇒ x = 2 × (a – x)
x = 2a/3

$$K_c = \frac{[C][D]}{[A][B]} = \frac{\frac{2a}{3} \cdot \frac{2a}{3}}{\frac{a}{3} \cdot \frac{a}{3}}$$

K_c = 4
Final Answer : (4)

51) **Explaining :** True statement among the following.
Concept : This question is based on active mass.
Solution :
(i) Active mass of pure liquids and solids are taken unity.
(ii) Active mass of gases can be expressed is terms of atm.
(iii) Law of mass action is applicable on reversible reaction
Final Answer : (4)

52) **Explaining :** The equilibrium constant (K_c) for the reaction HA + B ⇌ BH^+ + A^- is 100. If the rate constant for the forward reaction is 10^5, then rate constant for the backward reaction

is
Concept : This question is based on equilibrium constant (K_c)
Solution :
$K_c = \frac{K_f}{K_b}$
$100 = \frac{10^5}{K_b}$
$K_b = 10^3$
Final Answer : (2)

53) **Explaining :** True statement.
Concept : This question is based on Type of reaction.
Solution :
(1) In exothermic reaction, heat is evolved during reaction.
(2) Product is more stable then reactant.
Final Answer : (4)

54)

Explaining : comparision of pH of solution.
Concept : This question is based on hydrolysis of salt.
Solution :
$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$
Lower the pK_a, better the acid $HClO_4$ has highest acidic strength.
Final Answer : (2)

55)

Explaining : pK_a of weak acid (HA) and pK_b of weak base (BOH) are 3.2 and 3.4 respectively. The pH of their salt (BA) solution is
Concept : This question is based on pH of WBWA salt solution.
Solution :
$pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$
$= 7 + \left(\frac{1}{2} \times 3.2\right) - \left(\frac{1}{2} \times 3.4\right) = 6.9$
Final Answer : (4)

56)

Explaining : The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T(K). The correct option for the pH of dimethylammonium acetate solution is.
Concept : This question is based on pH of WBWA salt solution.
Solution :
$pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$

$$7+\left(\frac{1}{2}\times 4.77\right)-\left(\frac{1}{2}\times 3.27\right)=7.75$$

Final Answer : (3)

57)

Explaining : Match the correct pH of solution.
Concept : This question is based on pH of different solutions.
Solution :
(A) $pK_a + pk_b = 14$ for acid-base conjugate pair $pK_b = 14 - 6 = 8$
(B) $[H^+]_t = 10^{-8} + 10^{-7} = 11\times10^{-8}$; pH = 8–log11 = 8–1.04 = 6.96
(C) $pH=\frac{1}{2}pK_a-\frac{1}{2}\log C \Rightarrow pH=\left(\frac{1}{2}\times 4.7\right)-\frac{1}{2}\log 10^{-2}=3.35$
(D) $[H^+]=\frac{10^{-3}\times 10^{-5}}{2}$ = 0.5 × 10^{-3} ; pH = 4 – log5 = 4 – 0.7 = 3.3
Final Answer : (3)

58)

Explaining : What will be the pH of pure water at 60°C ?
Concept : This question is based on explanation of water.
Solution : pH decreases with increase in temperature.
Final Answer : (3)

59) **Explaining :** The K_a values of acetic acid, hypochlorous acid and formic acid are 1.74×10^{-5}, 3.0×10^{-8} and 1.8×10^{-4} respectively. Which of the following orders of pH of 0.1 mol dm^{-3} solutions of these acids is correct ? **Concept :** This question is based on pH of weak acid.
Solution : Higher the value of K_a, Higher the acidic strength.
Final Answer : (2)

60)

Explaining : K_a for CH_3COOH is 1.8 × 10^{-5} and K_b for NH_4OH is 1.8 × 10^{-5}. The pH of ammonium acetate will be
Concept : This question is based on hydrolysis of WBWA salt.
Solution :
$$pH=7+\frac{1}{2}pK_a-\frac{1}{2}pK_b$$
$$\Rightarrow 7+\left(\frac{1}{2}\times 4.74\right)-\left(\frac{1}{2}\times 4.74\right)$$
Final Answer : (3)

61) **Explaining :** Identify correct statement.
Concept : This question is based on common ion effect.
Solution : In presence of HCl, dissociation of H_2S decreases.
Final Answer : (2)

62)

Explaining : Identify the correct statement.
Concept : This question is based on hydrolysis of WBWA salt.
Solution : $pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$
Final Answer : (1)

63)

Explaining : At 25°C, the dissociation constant of a base, BOH is 1.0×10^{-12}. The concentration of hydroxyl ions in 0.01 M aqueous solution of the base would be:
Concept : This question is based on Ostwald's dilution law.
Solution : $[OH^-] = \sqrt{K_b \times C} = \sqrt{10^{-12} \times 10^{-2}} = 10^{-7}$
Final Answer : (2)

64)

Explaining : Aqueous solution of which of the following compounds is the best conductor of electric current ?
Concept : This question is based on electrolytes.
Solution : Strong electrolytes are good conductors.
Final Answer : (4)

65)

Explaining : What is the conjugate base of OH^- ?
Concept : This question is based on acid-base conjugate pair.
Solution : $OH^- - H^+ \longrightarrow O^{-2}$
Final Answer : (4)

66)

Explaining : There is transfer of five electrons.
Concept : This question is based on balancing of redox reaction
Solution : $\overset{+7}{Mn}O_4^- \rightarrow \overset{+2}{Mn}$
Final Answer : (4)

67)

Explaining : In the following reaction.
Concept : This question is based on types of redox reaction.
Solution : It is disproportionation in which phosphorus is oxidised as well as reduced.
Final Answer : (2)

68)

Explaining : Oxidation state of Cr
Concept : This question is based on oxidation number
Solution :
$[Cr(NH_3)_4Cl_2]^+$
x + 4(0) – 2 = +1
⇒ x = +3
Final Answer : (3)

69)

Explaining : Correct coefficients of the reactants
Concept : This question is based on balancing of redox reaction
Solution : -
$16H^+ + 2MnO_4^- + 10e^- \rightarrow 2Mn^{+2} + 8H_2O$
$5C_2O_4^{2-} \rightarrow 10CO_2 + 10e^-$

$16H^+ + 2MnO_4^- + 5C_2O_4^{2-} \rightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$
Final Answer : (2)

70)

Explaining : Value of n
Concept : This question is based on balancing of redox reaction
Solution :
$Cr_2O_7^{2-} + 14H^+ + nFe^{2+} \rightarrow 2Cr^{3+} + nFe^{3+} + 7H_2O$
–2+14 ⇒ (+12)+6×2 = 24 (+6)+6×3 ⇒ 24
Final Answer : (2)

71)

Explaining : Increasing order of oxidation number
Concept : This question is based on oxidation number
Solution :
$NH_4^+ < N_2O < NO < NO_2 < NO_3^-$
Oxidation states of nitrogen respectively are –3, +1, +2, + 4, + 5
Final Answer : (2)

72)

Explaining : Stoichiometric cofficient
Concept : This question is based on balancing of redox reaction
Solution : The balanced chemical equation is :
$3I_2 + 6OH^- \longrightarrow IO_3^- + 5I^- + 3H_2O$
p = 3, q = 6, r = 5
Final Answer : (4)

73) **Explaining :** Atomic parameters.

Concept : This question is based on atomic parameters.
Solution : Electron is not present in nucleus.
Final Answer : (4)

74) **Explaining :** Atomic parameters.
Concept : This question is based on atomic parameters.
Solution :
$m_n = 1.008$ amu ; $m_p = 1.007$ amu
$\frac{m_n}{m_p} > 1$
Final Answer : (2)

75) **Explaining :** Rutherford's experiment.
Concept : This question is based on rutherford's experiment.
Solution : (According to theory)
Final Answer : (3)

76)

Explaining : Rutherford's α-particle scattering experiment ?
Concept : This question is based on Rutherford's α-particle scattering experiment ?
Solution : The positive charge is concentrated in a very small volume of the atom.
Final Answer : (1)

77) **Explaining :** Angular momentum.
Concept : This question is based on Bohr's atomic model.
Solution : p – shell → n = 6 ⇒ mvr = $\frac{6h}{2\pi} = \frac{3h}{\pi}$
Final Answer : (1)

78)

Explaining : Atomic parameters.
Concept : This question is based on atomic parameters.
Solution :
A = 39 $(n_n = n_p + 1)$
$n_n + n_p = 39$
$n_p + 1 + n_p = 39$
⇒ $n_p = 19$ ⇒ $n_{e^-} = 19$
$n_n = 20$
Final Answer : (1)

79)

Explaining : Energy of e^-.
Concept : This question is based on Bohr's atomic model.

Solution : $E = -13.6 \times \frac{z^2}{n^2} = -13.6 \times \frac{(1)^2}{(2)^2} = -3.4$ eV
Final Answer : (2)

80)

Explaining : Kinetic energy.
Concept : This question is based on kinetic energy.
Solution : as r↑ K. E↓ ⇒ x > y
Final Answer : (2)

81) **Explaining :** Spectrum.
Concept : This question is based on spectrum.
Solution :

$$\frac{(\nu_{max})_{Lyman}}{(\nu_{max})_{Brackett}} = \frac{\left[\frac{1}{(1)^2} - \frac{1}{\infty}\right]}{\left[\frac{1}{(4)^2} - \frac{1}{\infty}\right]}$$

$= \frac{16}{1} \Rightarrow 16 : 1$
Final Answer : (2)

82)

Explaining : No. of moles.
Concept : This question is based on No. of moles.
Solution : $n_{O_3} = \frac{12.044 \times 10^{22}}{N_A} = 0.2$ mol
Final Answer : (4)

83)

Explaining : Mole fraction.
Concept : This question is based on mole fraction.
Solution :
$n_A = 5$ mol
$n_B = 15$ mol
$\Rightarrow x_A = \frac{5}{5+15} = \frac{5}{20} = \frac{1}{4} = 0.25$
Final Answer : (3)

84)

Explaining : Stoichiometry.
Concept : This question is based on stoichiometry.
Solution :
1L 3L

$N_2 + 3H_2 \rightarrow 2NH_3$
$3L \rightarrow 3 \times 3 = 9L$ of H_2
Final Answer : (3)

85)

Explaining : Stoichiometry
Concept : This question is based on stoichiometry.
Solution :
(Concept of L.R)
ratio A is in excess $\rightarrow \frac{5}{2} < \frac{3}{6} \rightarrow$ B is L.R
Final Answer : (1)

86) **Explaining :** Percentage of atom in a compound.
Concept : This question is based on percentage of atom in a compound.
Solution : % of H = $\frac{4}{16} \times 100 = 25\%$
Final Answer : (1)

87)

Explaining : Empirical formula.
Concept : This question is based on empirical formula.
Solution :
$$\underset{A}{\frac{9 \times 10^{20}}{N_A}} : \underset{B}{\frac{3 \times 10^{20}}{N_A}}$$
$\Rightarrow 3 : 1 \Rightarrow A_3B$
Final Answer : (2)

88)

Explaining : Molality.
Concept : This question is based on molality.
Solution :
$$m = \frac{(moles)_{solute}}{(w_{solvent})_{kg}}$$
$$= \frac{2}{20} = 0.1 \text{ m}$$
Final Answer : (1)

89)

Explaining : Energy of photon.
Concept : This question is based on Bohr's electronic model.

Solution : $\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = 2$
Final Answer : (2)

90)

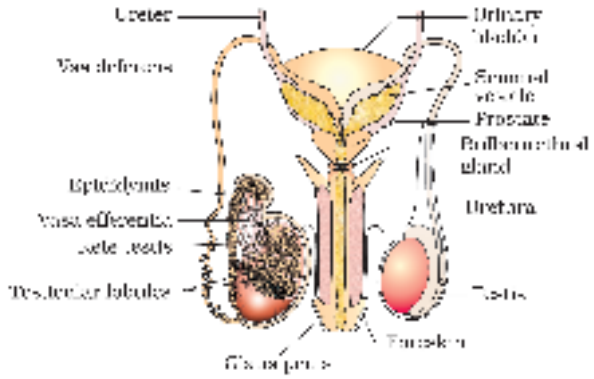
Explaining : Angular momentum.
Concept : This question is based on Bohr's theory.
Solution : $L = \frac{nh}{2\pi} = \frac{5h}{2\pi} = 2.5\frac{h}{\pi}$
Final Answer : (4)

BIOLOGY

91) **Question Explanation:** This question is asking about the gland whose secretion lubricates the penis.
Concept : Human Reproduction – Male Accessory Glands.
Solution: The bulbourethral (Cowper's) glands secrete a clear mucus that lubricates the penis and urethra during sexual activity.
Final Answer: (3)

92) **Solution :** Scrotum helps to maintains the temperature lower than that of body. Its essential for spermatogenesis.
Final Answer : (3)

93) **Solution :**



Final Answer : (3)

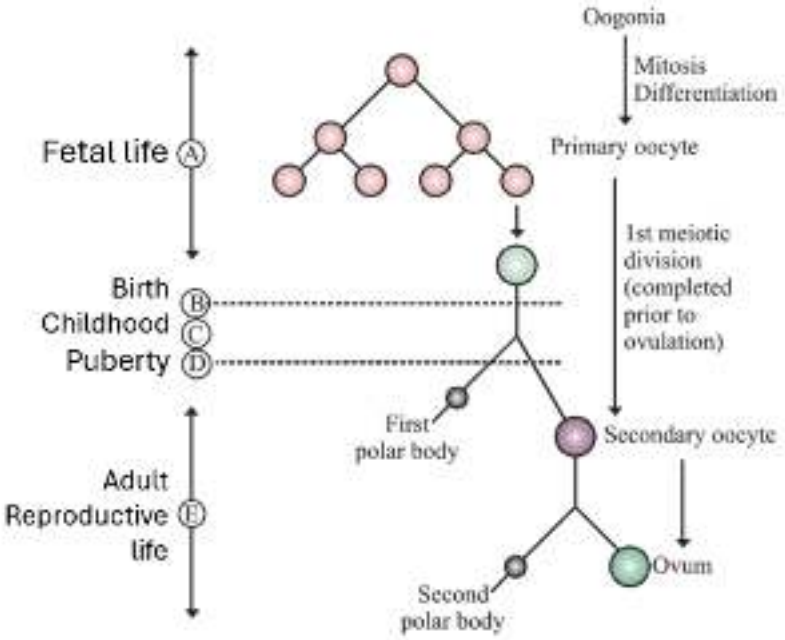
94) **Solution :** Path of sperm = Rete testis → Epididymis → Vasa deferens → Urethra
Final Answer : (4)

95) **Solution :** Relaxin hormone is secreted by both placenta and ovary (Corpus luteum).
Final Answer : (2)

96) **Explain Question :** to identify A and B and their respective functions

Concept : This question is based on embryonic development.
Solution :
A – Trophoblast, whose function is to Attach to the endometrium
B – Inner cell mass, which will Differentiated as embryo
Final Answer : (1)

97) **Explain Question :** The following refers to schematic representation of oogenesis. Identify A to E correctly.
Concept : This question is based on Oogenesis
Explanation :



Final Answer : (2)

98) **Solution :** I → III → V → IV → II
Final Answer : (2)

99)

Explain Question : Identify the four statements (a–d) and select the option which includes all the incorrect ones only.

Concept : Male reproductive system

Solution : The correct option is (2).

(a) Incorrect — the number “750 tubules per lobule” is wrong (each lobule has only 1–3 seminiferous tubules).

(b) Incorrect — Sertoli cells are supportive cells and do not undergo meiosis.

(c) Correct — interstitial spaces do contain Leydig cells and blood vessels.

(d) Correct — scrotum helps maintain lower temperature for spermatogenesis.

Final answer : (2) Statements (a) and (b).

100) **Solution :** Progesterone and estrogen are produced in women throughout a woman's regular menstrual cycle.
Final Answer : (4)

101) **Solution :** (i) – c, (ii) – e, (iii) – a, (iv) – b, (v) – d
(i) End of one month → (c) Heart formation
(ii) End of second month → (e) Limbs and digits formation
(iii) End of three month → (a) Most of the major organ systems developed
(iv) During fifth month → (b) Appearance of hair on the head (v) End of sixth month → (d) Eye-lids separated **Final Answer : (4)**

102) **Solution :** Structures present only in male as pairs are testes epididymis, vas deferens and seminal vesicles.
Final Answer : (3)

103) **Explain Question :** to match the column correctly
Concept : This question is based on Human reproduction.
Solution : A - iii, B - iv, C - i, D - ii
(A) Menopause → (iii) 50 years of age
(B) Fertilization → (iv) Ampullary region of fallopian tube
(C) Androgens → (i) Leydig Cells
(D) Oxytocin → (ii) Parturition
Final Answer : (2)

104)

Question Explanation : The question asks what the ruptured Graafian follicle develops into after ovulation.
Concept : Menstrual cycle
Solution : The correct answer is option 3.

A. After ovulation, the mature Graafian follicle ruptures and releases the secondary oocyte. The remaining parts of the follicle then transform into a temporary endocrine structure called the corpus luteum, which produces hormones to support a potential pregnancy.

Final Answer : (3)

105) **Solution :** Functional mammary gland is the characteristic feature of all female mammals.
Final Answer : (3)

106)

Explain Question : to identify the day of menstrual cycle, on which the thickness of uterine

wall is maximum.

Concept : This question is based on menstrual cycle.

Solution : the thickness of uterine wall is maximum around 23 rd day of menstrual cycle.

Final Answer : option (4)

107)

Question Explanation : Calculate number of meiotic divisions needed to produce 300 sperms and ova.

Concept : The question is based on gametogenesis (spermatogenesis and oogenesis).

Solution : The correct answer is option 1.

A. In spermatogenesis, 1 primary spermatocyte → 4 sperms after meiosis.
   So, for 300 sperms → 300÷4=75 meiotic events.

B. In oogenesis, 1 primary oocyte → 1 ovum (rest are polar bodies).
   So, for 300 ova → 300 meiotic events.

Final Answer : (1)

108) **Explain Question :** to Match the Column–A with Column–B
Concept : Menstrual cycle
Explanation : A–(ii), B–(iii), C–(iv), D–(i)
(A)Follicular phase → (ii) Development of follicle
(B) Luteal phase → (iii) Formation of corpus luteum
(C)Menstrual phase → (iv) Breakdown of endometrium
(D) Ovulatory phase → (i) Release of secondary oocyte
Final Answer : (3)

109) **Solution :** Bartholin glands → Secrete a clear, thick mucoid fluid
Their primary function is to provide vaginal and vulval lubrication during sexual arousal to reduce friction and make contact more comfortabel.
Final Answer : (4)

110)

Question Explain : Evaluate the truth and relationship between the assertion and reason regarding implantation.

Concept: Implantation

Solution: The correct answer is option 4.

A. Assertion (A) is true: Implantation is the process where the blastocyst attaches to and embeds into the endometrial lining.

B. Reason (R) is also true: Adhesive interactions between the trophoblast (outer layer of blastocyst) and endometrium are crucial for successful implantation.

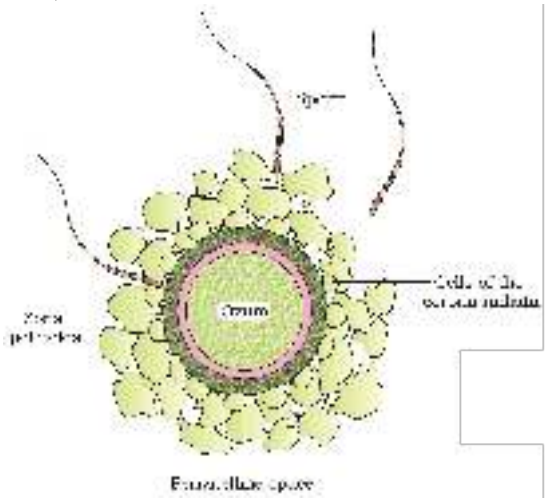
C. Since the adhesive nature directly facilitates the embedding process, R correctly explains A.

Final Answer: 4

111) **Explain Question :** Check correctness of statements about hormones in the menstrual cycle.
Concept : Menstrual cycle
Solution :
Statement I : True → During the follicular phase, FSH & LH (gonadotropins) rise gradually, stimulating follicle growth.
Statement II : True → When progesterone drops (due to degeneration of corpus luteum), the endometrium is shed, causing menstruation.
Final Answer : (4)

112) **Solution :** The assertion is correct; the perimetrium is a thin, membranous layer. The reason is also correct; it is the outer layer of the uterus. However, the reason does not explain why it is a thin membranous layer.
Final Answer : (2)

113) **Solution :**



Final Answer : (2)

114)

Question explanation : Identify which structures contribute to the formation of seminal plasma.

Concept : Male accessory glands.

Solution :

A. Sertoli cells (a) → nourish sperms, do not form seminal plasma

B. Seminal vesicle (b) → major contributor to seminal plasma

C. Spermatogonia (c) → sperm-forming cells, not plasma

D. Leydig cells (d) → secrete testosterone, not plasma

E. Bulbourethral gland (e) → contributes mucus to seminal plasma

F. Prostate gland (f) → contributes alkaline fluid to seminal plasma

So, seminal plasma is formed by seminal vesicles, bulbourethral glands and prostate gland.

Final answer : Option 4

115) **Solution :** All statements are correct regarding Breast (Mammary gland).
Final Answer : (3)

116)

Explain Question : The question asks about the nuclear content and structure of human sperm.

Concept : This question is based on the structure of sperm.

Solution : The correct answer is option 3.

A. A sperm cell, being a male gamete, contains a **haploid nucleus** which is located in its **head region**. The middle piece, not the tail, contains numerous mitochondria to provide energy for movement.

Final answer : option (3)

117)

Question Explanation : Identify when the first polar body is released during oogenesis.

Concept : This question is based on Oogenesis.

Solution : The correct answer is option 2.

Option 2 – After completion of meiosis-I: Correct.

A. In oogenesis, primary oocyte completes meiosis-I at puberty prior to ovulation, forming secondary oocyte and first polar body.

Option 1 – Before birth: Incorrect.

A. Primary oocytes are formed before birth but are arrested in prophase-I; no polar body is formed yet.

Option 3 – At the time of fertilizations: Incorrect.

A. Fertilization triggers completion of meiosis-II, forming the second polar body.

Option 4 – At the time of formation of haploid ovum: Incorrect.

A. This occurs after meiosis-II, not the first polar body.

Final answer : (2)

118)

Question Explanation: The question asks which group of hormones is produced only during pregnancy.

Concept: Placenta

Solution: Hormones produced specifically during pregnancy(by placenta mainly) include HPL (Human Placental Lactogen), HCG (Human Chorionic Gonadotropin), and Relaxin. These hormones are secreted by the placenta to support fetal growth, maintain the corpus luteum, and prepare the birth canal respectively. Other hormones like prolactin, progesterone, thyroxine, and cortisol are present in non-pregnant states too.

Final answer: Option 2

119)

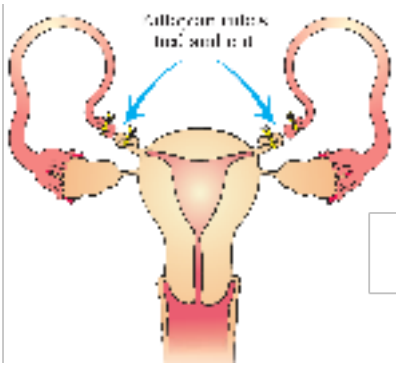
Question explanation : Identify the hormone released due to foetal ejection reflex during childbirth.
Concept : This question is based on Parturition.
Solution : The correct answer is option 4.

A. The foetal ejection reflex, triggered by the fully developed foetus and placenta, stimulates the

maternal pituitary to release oxytocin, which induces strong uterine contractions for childbirth.

Final answer : option (4)

120) **Solution :**



Final Answer : (1)

121) **Solution :** Copper–T → Uterine cavity
Final Answer : (3)

122) **Question Explanation:** The question asks which of the listed methods are natural contraceptive methods.
Concept: Natural contraceptive methods.
Solution: Periodic abstinence (A) involves avoiding intercourse during the fertile period,which is a natural method
Coitus interruptus (B) is withdrawal before ejaculation, which is a natural method but less reliable.
Lactational amenorrhoea (C) is a natural method where breastfeeding prevents ovulation.
Final Answer: (3)

123) **Question Explanation:** The question is asking which complications can arise from sexually transmitted diseases (STDs) if they are not diagnosed and treated in time.
Concept: Assisted Reproductive Technologies (ART).
Solution: Untreated STDs can lead to pelvic inflammatory disease (PID), infertility, ectopic pregnancy, and may cause stillbirth or congenital diseases in newborns. Hence, all listed conditions can result from untreated STDs.
Final Answer: (4)

124) **Explain Question :** Match the following and give the answer for the correct match.
Concept : Assisted Reproductive Technologies (ART)
Explanation : A-iii B-i C-iv D-ii
ZIFT → Zygote up to 8 blastomere stage transferred into fallopian tube → iii
IUT → More than 8 blastomere stage transferred into uterus → i
GIFT → Transfer of ovum collected from ovary into fallopian tube → iv
ICSI → Sperm directly injected into ovum cytoplasm → ii
Final Answer : (4)

125) **Explain Question :** to identify the Weekly oral pills for females
Concept : Hormonal contraceptive pills
Explanation : Saheli is a non-steroidal weekly oral contraceptive pill
Final Answer : (3)

126)

Explain Question : to identify what will happen if an emergency contraceptive pill is given to a female who is 2 weeks pregnant.

Concept : This question is based on contraceptive methods.

Solution : At 2 weeks of pregnancy, implantation has already happened, and the embryo is starting to develop. Emergency contraceptive pills (ECPs) work primarily by preventing ovulation or fertilization. So here ECPs will not have any effect, Pregnancy is unaffected.

Final Answer : (3)

127) **Question Explanation:** Asking about a mathod not suitable for population control.
Concept : Temporary contraceptives & MTP.
Solution : Allmost it can terminote the pregnency but M.T.P. is not considered as ethical option for population contral.
While the other contra ceptives can be used as Temporary contraceptives for population control.
Final Answer : option (4).

128)

Question Explanation: Natural methods of contraception include the following ?

Concept: Natural contraception methods

Solution: The correct matching is option 1
Natural methods = no device/chemical, rely on timing or behaviour.
Only coitus interruptus, periodic abstinence, and lactational amenorrhea are natural.
IUD and diaphragm are NOT natural (they are mechanical/barrier methods).

Final Answer: Coitus interruptus, periodic abstinence

129)

Explain Question : The contraceptive method with almost no side effect is :-
Concept : This question is based on birth control methods
Explanation : Periodic abstinence (safe period method) is a natural contraceptive method and has almost no side effects.
Final answer : (1)

130)

Question Explanation: Identify which method has been used for emergency contraception.
Concept: Emergency contraception

Solution: The correct answer is option 4

A. Emergency contraception is used to prevent pregnancy after unprotected sexual intercourse or contraceptive failure. High-dose oral contraceptives containing progesterone and estrogen are commonly used for this purpose. They work mainly by inhibiting or delaying ovulation and, in some cases, preventing implantation.

   Other options like diaphragm, daily tablets, or Saheli are either regular contraceptive methods or barrier methods and are not used specifically for emergency contraception.

   Final Answer: Option 4

131) **Solution :** MTP is safe during upto 12 weeks of pregnancy.
Final Answer : (2)

132)

Question Explanation : Identify where embryos of different cell stages are transferred after IVF.

Concept : The question is based on assisted reproductive technologies .

Solution : The correct answer is option 1.

A. Embryos with up to 8 cells are usually transferred into the fallopian tube (ZIFT).

B. Embryos with more than 8 cells are transferred into the uterus (IUT).

So, statement given in option (1) correctly states that embryos with more than 8 cells are implanted in the uterus.

Final answer : Option 1

133) **Question Explanation :** The question is asking about the growth rate of population in India.
Concept : Population growth rate
Solution : India’s population growth rate is less than 2% per year. Although the population is large, the annual growth rate has declined due to factors like increased awareness, family planning, and improved education.
Final Answer : (2)

134) **(1) Correct:** Progestogens, alone or in combination with estrogen, are used as oral contraceptives.
(2) Correct: Saheli is a **"once-a-week"** non-steroidal oral contraceptive pill.

(3) Correct: Most **oral contraceptive pills** are taken **daily for 21 days,** followed by a 7-day gap.
(4) Incorrect: Saheli has low side effects and high contraceptive value, making this statement wrong.
Thus, the incorrect statement is **Option 4.**

135) **Solution :** If the vas-deferens of a man are surgically cut or blocked then semen will be without sperms.
Final Answer : (1)

136) **Question Explanation:** Identify the correct world population figures for the years 1900 and 2000.
Concept: Population growth.
Solution:
A = 2 billion → World population in 1900 was approximately 2 billion.
B = 6 billion → By 2000, global population had risen to around 6 billion.
Final Answer: (2)

137) **Solution :** Saheli is a non-steroidal preparation taken once a week.
Final Answer : (2)

138) **Solution :** CuT, Cu7, and Multiload 375 are examples of copper releasing IUDs.
Final Answer : (2)

139) **Question Explanation:** Identify a contraceptive device and its correct site of implantation in a woman.
Concept: Birth control methods
Solution:

A. Implants are small rods or capsules that release hormones (usually progestin) to prevent pregnancy.
B. They are implanted subcutaneously, typically in the inner side of the upper arm, where they continuously release the hormone over months or years.

Final Answer: (3)

140)

Question Explanation : identify correctness of assertion and reason about hymen as an indicator of virginity.
Concept : This question is based on Female external genitalia.
Solution : The correct answer is option 1.

A. Assertion is correct because hymen is not a reliable indicator of virginity.
B. Reason is also correct. Hymen may persist after coitus and can tear due to non-sexual

activities like sports, exercise, or sudden fall.

C. (R) is correct explanation of (A) because if hymen can persist even after sex or break without sex, its condition cannot reliably prove virginity, directly supporting the assertion.

Final Answer : option (1)

141) **Solution:** Statements I, II, and III are correct.
Statement IV is incorrect because the generative cell is small and floats in the cytoplasm of the vegetative cell (which is larger, has abundant food reserve, and possesses a large irregularly shaped nucleus). The generative cell is spindle-shaped with dense cytoplasm and a nucleus.
Final Answer: (1)

142) **Solution:**
Statement I is correct (MMC differentiates in the micropylar region).
Statement II is incorrect because monosporic development involves three sequential mitotic (free-nuclear) divisions of the functional megaspore, not meiotic.
Statement III is correct (typical Polygonum type embryo sac is 7-celled and 8-nucleate).
Final Answer: (2)

143) **Solution:** All three statements are factually accurate. Geitonogamy is ecologically cross-pollination but genetically autogamy. Dioecy (male and female flowers on separate plants) prevents both self-pollination types. Self-incompatibility is a genetically controlled mechanism to prevent inbreeding.
Final Answer: (4)

144) **Solution:** Monoecious plants have separate male and female flowers on the same individual plant. Because a single flower cannot self-pollinate (no autogamy), but pollen can move between different flowers on the same plant (geitonogamy), both Assertion and Reason are true and correctly linked.
Final Answer: (1)

145) **Solution:** Embryo development requires nutrition. Therefore, endosperm forms first to ensure a nutrient supply for the zygote's subsequent cleavage and differentiation.
Final Answer: (4)

146) **Solution:** Apomixis creates clones of the mother plant without meiosis or syngamy; hence, apomictic offspring do not segregate traits and remain genetically uniform (no variable offspring). Thus, Assertion is false, but Reason is true.
Final Answer: (3)

147) **Solution:** Tapetum is the innermost nourishing layer of the microsporangium. Nuclear division without cytoplasmic division leads to the multinucleate condition and high metabolic activity required for nourishing microspores.
Final Answer: (1)

148) **Solution:** Ovary wall turns into Pericarp; Persistent nucellus forms Perisperm (e.g., Black pepper, Beet); Ovule becomes Seed; Integuments turn into Seed coat (Testa & Tegmen).
Final Answer: (1)

149) **Solution:** Bright colors, nectar, and scent (3) are traits of entomophilous (insect-pollinated) flowers. Wind pollination relies on light/non-sticky pollen, exposed stamens, feathery stigmas, and simple ovaries.
Final Answer: (2)

150) **Solution:**

A. Synergids (n), individual Polar nuclei (n), and Functional Megaspore (n) are haploid.
B. The Central cell contains two polar nuclei, making its nuclear content n+n.
C. PEN is triploid (3n) formed post-triple fusion.

Final Answer: (1)

151) **Solution:** The 4 wall layers of a microsporangium from outside to inside are:
Epidermis → Endothecium → Middle layers → Tapetum.
Final Answer: (1)

152) **Solution:**
P is the Chalazal pole (basal part).
Q is the Nucellus (tissue with reserve food materials).
R is the Micropyle (pore at the micropylar end).
S is the Funicle (stalk of the ovule; Hilum is the junction where funicle fuses with ovule body).
Final Answer: (2)

153) **Solution:**
$2n = 24 \Rightarrow n = 12$.
Aleurone layer is part of triploid endosperm $(3n) = 3 \times 12 = 36$.
Synergid is haploid $(n) = 12$.
Integument is maternal diploid $(2n) = 24$.
Final Answer: (1)

154) **Solution:** *Eichhornia* (Water hyacinth) and *Nymphaea* (Water lily) are aquatic, but their flowers emerge above the water surface and are pollinated by insects or wind. In *Vallisneria*, female flowers reach the surface by long stalks. In *Zostera*, pollen grains are long/ribbon-like and carried submerged inside water.
Final Answer: (3)

155) **Solution:** Nucellar polyembryony (a form of adventive embryony) occurs when maternal nucellar cells (2n) surrounding the embryo sac start dividing and protrude into the embryo sac to develop into embryos.
Final Answer: (2)

156) **Solution:** The filiform apparatus consists of cellular thickenings at the micropylar tip of

synergids that secrete chemo-attractants to guide the entry of the pollen tube.
Final Answer: (2)

157) **Solution:** Endothecium cells feature fibrous bands of α-cellulose. They lose water at maturity, creating mechanical tension that causes the anther to rupture along the stomium.
Final Answer: (2)

158) **Solution:**
MMC (2n) = 18 ⇒ n = 9.
Endosperm (3n) = 3 × 9 = 27.
Generative cell (n) = 9.
Final Answer: (1)

159) **Solution:** Cryopreservation of pollen grains is carried out in liquid nitrogen at –196°C for long-term storage in crop breeding programs.
Final Answer: (1)

160) **Solution:** Coconut water from tender coconut is free-nuclear endosperm (made of thousands of free nuclei), while the surrounding white flesh (kernel) is cellular endosperm.
Final Answer: (1)

161) **Solution:** Integument (2n), Megaspore Mother Cell MMC (2n). Microspore is formed by meiosis (n), and male gametophyte is formed by mitotic divisions of the microspore (n).
Final Answer: (1)

162) **Solution:** Chasmogamous flowers are normal open flowers with exposed stamens and pistils, whereas cleistogamous flowers remain unopened.
Final Answer: (2)

163) **Solution:** Application of low concentrations of Auxins and Gibberellins induces fruit development without fertilization (parthenocarpy), producing seedless commercial fruits (e.g., seedless grapes, tomatoes).
Final Answer: (2)

164) **Solution:** Perisperm is the residual, persistent nucellus present in mature seeds of black pepper, beet, and castor.
Final Answer: (2)

165) **Solution:** The single Cotyledon of monocots (poaceae/grasses) is situated laterally and is termed **Scutellum**.
Final Answer: (2)

166) **Solution:** Coleorhiza protects the radicle/root cap, while Coleoptile encloses the plumule/shoot apex.

Final Answer: (2)

167) **Solution:** Apomixis preserves hybrid traits without segregation because offspring are clonal. Farmers can save and sow apomictic seeds season after season without loss of hybrid vigor.
Final Answer: (2)

168) **Solution:** In ~60% of angiosperms, pollen grains are shed at the 2-celled stage (1 vegetative cell + 1 generative cell). In the remaining ~40%, the generative cell divides mitotically to yield 2 male gametes before shedding (3-celled stage).
Final Answer: (2)

169) **Solution:** Antipodals (3 cells at the chalazal end) degenerate after fertilization.
Final Answer: (2)

170) **Solution:** Pea, Groundnut, and Beans are dicot non-albuminous seeds where endosperm is completely consumed prior to seed maturation. Castor, Maize, Wheat, and Barley are albuminous (endospermic) seeds.
Final Answer: (1)

171) **Solution:** Unisexual female flowers lack stamens/anthers completely, so emasculation is unnecessary. Only bagging is required before stigma receptivity.
Final Answer: (3)

172) **Solution:** Apple, Pear, and Strawberry are false fruits (pseudocarps) where the swollen thalamus becomes the edible fleshy fruit body.
Final Answer: (2)

173) **Solution:** Wheat and rice pollen loses viability within 30 minutes of release. Members of Rosaceae, Leguminosae, and Solanaceae retain pollen viability for months under appropriate conditions.
Final Answer: (1)

174) **Solution:** Double fertilization consists of:
1. **Syngamy:** Male gamete (n) + Egg cell (n) → Zygote (2n)
2. **Triple Fusion:** Male gamete (n) + 2 Polar nuclei (n+n) → PEN (3n).

Final Answer: (1)

175) **Solution:** The standard sequence is:
1. Emasculation of bisexual bud before anther dehiscence.
2. Bagging to prevent contamination by unwanted pollen.
3. Dusting mature desired pollen when stigma becomes receptive.
4. Re-bagging until fruit matures

Final Answer: (1)

176) **Solution :** The hilum is a scar on the ovule, where funicle is attached.
Hence 4th option is correct.
Final Answer : (4)

177)

The question asks to match the terms in **Column-l** with the correct descriptions in **Column-ll.** Here are the correct matches:

A. **(a) Pistils fused together:** This describes **syncarpous**, where multiple pistils fuse into a single structure (iii).
B. **(b) Formation of gametes:** This refers to **gametogenesis**, the process of producing gametes (i).
C. **(c) Hyphae of higher Ascomycetes:** These hyphae are typically **dikaryotic**, meaning each cell has two distinct nuclei (iv).
D. **(d)** A typical anther is bilobed**.**

Final Answer : (2)

178) **Solution :** The given figure shows wind pollinated plant inflorescence showing flowers with well exposed stamens.
Final Answer : (1)

179) **Explanation:** The given question is a multiple-choice question about angiosperms, specifically testing your knowledge of their reproductive structures and processes.

A. **(a) A microsporophyll contains stigma, style and ovary. - Incorrect.** Microsporophylls are male reproductive structures that produce pollen grains. They do not contain female structures like stigma, style, and ovary.
B. **(b) All megaspores produced from megaspore mother cell take part in embryo sac formation. - Incorrect.** Only one megaspore survives and undergoes meiosis to form the embryo sac.
C. **(c) Hilum is the place where ovule joins with funicle. - Correct.** The hilum is the scar on the ovule where it was attached to the funicle.
D. **(d) Basal part of an ovule is represented as micropylar end. - Incorrect.** The micropylar end is the opposite end of the hilum, where the pollen tube enters the ovule.
E. **(e) Perisperm is present in black pepper. - Correct.** Perisperm is a persistent nucellus that serves as a storage tissue in some seeds, including black pepper.

Therefore the given answer is **correct.** Only statements (c) and (e) are true.
Final Answer: (1)

180) **Explanation:**

A. The embryo sac in dicots typically has 7 cells, and the 8 nuclei are arranged as follows:
B. 3 antipodal cells with 1 nucleus each.
C. 2 synergid cells, each with 1 nucleus.
D. 1 egg cell with 1 nucleus.
E. 1 central cell with 2 nuclei (which later fuse to form the primary endosperm nucleus).

Thus, the arrangement is **3 + 2 + 3**.

Correct answer: 2. 3 + 2 + 3.